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G. Lehmann
Telecom SudParis
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# The Incident Detection Message Exchange Format version 2 (IDMEFv2)

## draft-lehmann-idmefv2-08

## Abstract

The Incident Detection Message Exchange Format version 2 (IDMEFv2) defines a data representation for security incidents detected on cyber and/or physical infrastructures.

The format is agnostic so it can be used in standalone or combined cyber (SIEM), physical (PSIM) and availability (NMS) monitoring systems. IDMEFv2 can also be used to represent man made or natural hazards threats.

IDMEFv2 improves situational awareness by facilitating correlation of multiple types of events using the same base format thus enabling efficient detection of complex and combined cyber and physical attacks and incidents.

This draft is maintained by the IDMEFv2 Task Force. Please consult our website for more information: https://www.idmefv2.org.

If approved this draft will obsolete RFC4765.

## Status of this Memo

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# 1. Introduction

The Incident Detection Message Exchange Format (IDMEF) is intended to solve the problem of security monitoring compartmentalization by proposing a single format to represent any type of incident, whether cyber or physical, intentional or accidental, natural or man-made.

Indeed security is often associated to the Confidentiality-Integrity-Availability triad, performance and availability management systems are still run independently from security management systems.

Additionally, with the adoption and integration of Internet of Things (IoT) and Industrial Internet of Things (IIoT) devices, and the exponential emergence of smart systems (transport, cities, buildings, etc), an increasingly interconnected mesh of cyber-physical systems (CPS) has emerged. This expansion of the attack and incident surfaces blurs the once-clear functions of cybersecurity and physical security.

Finally, as IT infrastructure moves out of data centers it becomes more exposed to external threats, including natural and man-made hazards.

Incident detection systems have traditionally focused on detecting cyber incidents or physical incident or availability incidents. There is an increasing need nowadays to have a unified view and management of all those incidents and their interconnection.

To achieve this goal the Incident Detection Message Exchange Format offers a unique data representation for multiple types of events:

- Cyber-security events (e.g. authentication failure/success, virus/malware detection, bruteforce/scan detection, etc.)
- Physical security events (e.g. intrusion detection, object detection, face or activity recognition, fire/smoke/noise/rain detection, etc.)
- Availability/observability/performance events (e.g. system failure, service malfunction, performance decrease, etc.)
- Natural and man made hazards events (e.g. wildfires, avalanches, droughts, earthquakes, pollution, fire, explosion, etc.)

## 1.1. IDMEFv2 deployment architecture

IDMEFv2 can be used to exchange incident detection information between specialized managers (SIEM, PSIM, NMS) and a universal "Cyber & Physical SIEM" (CPSIEM) or directly from specialized analyzers and a CPSIEM.

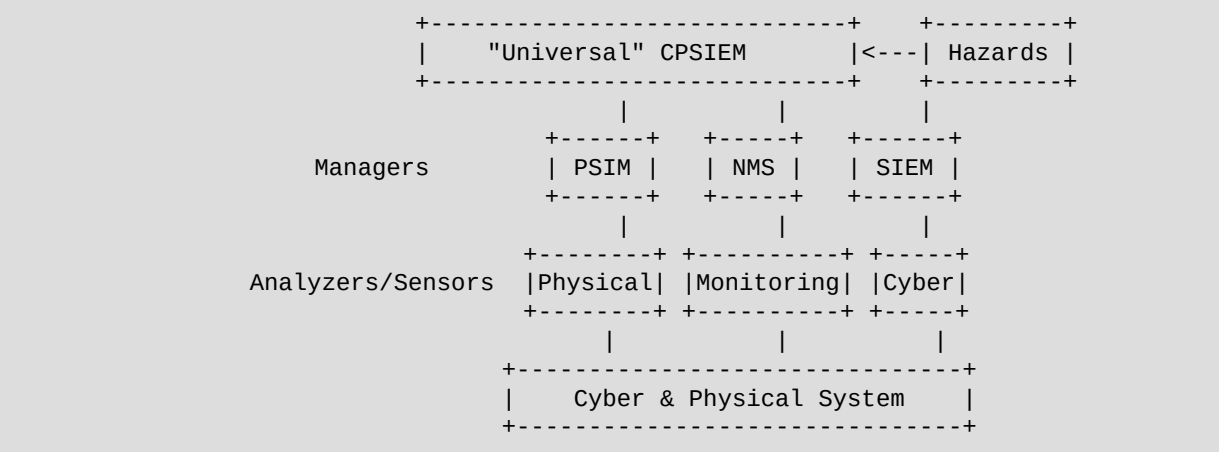



Figure 1: IDMEF Use Architecture

IDMEFv2 improves situational awareness by enabling correlation of multiple types of events using the same base format.

This document defines a model serialization methods for the purpose of describing and sharing these events.

## 1.2. IDMEFv1 (Intrusion Detection Message Exchange Format) - RFC 4765 - Legacy

IDMEFv2 (Incident Detection Message Exchange Format) is based on IDMEFv1 (Intrusion Detection Message Exchange Format) concepts. But IDMEFv1 was cyber intrusion focused as IDMEFv2 perimeter is much larger. Thus retro-compatibility although partly possible has not been a priority.

## 1.3. Relationship between IDMEFv2 and other event/incident formats

IDMEFv2 focuses essentialy on high level event/incident correlation and detection.There are many standard and proprietary formats on the incident detection market and in particular on the cybersecurity market. IDMEFv2 is complementary to most of these formats.

IDMEFv1 (Intrusion Detection Message Exchange Format - RFC 4765) : IDMEFv2 (Incident Detection) replaces and obsoletes IDMEFv1 (Intrusion Detection) by covering a wider spectrum.

IODEFv2 (Incident Object Definition Exchange Format - RFC 5070) : IDMEFv2 helps detect incident. When an incident is detected it will be analysed and eventually fully described and shared with other security teams through IODEFv2. IDMEF is used upstream IODEFv2. IDMEFv2 alerts can be “attached” to IODEFv2 object to provide technical details about incidents.

Syslog (System Logging) : Syslog is a lossy format with no formal structure. Syslog can be used by sensors to send information to analyzers. Out of those multi-format syslogs the analyzer might detect an incident or an event of interest. The analyzer will then use IDMEFv2 to notify the manager which might correlate this information with other data to confirm the incident.

SNMP (Simple Network Management Protocol) : SNMP polls information from devices which is then compared to thresholds to detect incident. IDMEFv2 can be used when incident is detected downstream of SNMP to communicate the incident to the manager. IDMEFv2 can have a similar role as SNMP Traps.

STIX (Structured Threat Information Expression) : is a language and serialization format used to exchange cyber threat intelligence (CTI). IDMEFv2 can help detect incidents which might lead to the creation and sharing of STIX information. Cyber analyzer can also rely on STIX information to detect incidents that will be notified in IDMEFv2 format.

OCSF (The Open Cybersecurity Schema Framework) is an open-source, vendor-agnostic standard designed to normalize security telemetry from diverse tools. It provides a common language and consistent structure for security event data, simplifying data ingestion, correlation, and analysis. OCSF can be seen as a “super” syslog to describe events before IDMEFv2 extract “incidents”. OCSF is limited to cyber security.

SIEM proprietory formats (CEF, LEEF, ECS, CIM, ...) : By covering cyber, physical and monitoring incidents type, IDMEFv2 offers a wider spectrum than those formats. Gateways between IDMEFv2 and those formats can be developped to connect legacy cyber detection systems to an IDMEFv2 architecture.

## 1.4. Existing Deployments and Adoption

IDMEFv2 is not a theoretical proposal. It has been developed, validated, and deployed within the framework of eight large-scale European research projects, funded by the Horizon 2020 and Digital Europe programmes. These projects — namely 7SHIELD, PRECINCT, CyberSEAS, ATLANTIS, ENDURANCE, KINAITICS, TESTUDO, and SAFE4SOC — address critical sectors such as space systems, energy grids, transportation, and government infrastructure. These implementations span multiple domains, including Security Operations Centers (SOCs), Physical Security Information Management (PSIM) systems, and critical infrastructure protection pilots.

This document aims to formalize this existing practice as an IETF Experimental RFC, to ensure interoperability, gather broader community feedback, and provide a stable foundation for future developments.

# 2. Terminology

## 2.1. Keywords

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in [RFC2119].

## 2.2. Normative sections

Implementations of IDMEFv2 are REQUIRED to fully implement:

- The data types defined in Section 3
- The data model defined in Section 5
- The JavaScript Object Notation (JSON) serialization method Section 5.8.

## 2.3. Concepts related to event processing

### 2.3.1. Event

An event is something that triggered a notice. Any incident starts off as an event or a combination of events, but not all events result in an incident. An event does not need be an indication of wrongdoing. E.g. someone successfully logging in or entering a building is an event.

### 2.3.2. Incident

An incident is an event that compromises or has a significant probability of compromising at least one of the organization's security criteria such as Confidentiality, Integrity or Availability. An incident may affect a production tool, personnel, etc. It may be logical, physical or organizational in nature. Last but not least, an incident may be caused on purpose or by accident.

### 2.3.3. Alert

An alert is a notification/message that a particular event/incident (or series of events/incidents) has occurred.

### 2.3.4. Manager

The manager is the central console toward which all analyzers send their alerts. The manager collects, correlates, stores and displays the alerts to the operators.

Example : - A SIEM (Security Information & Event Management) or a Log Manager) - A PSIM (Physical Security Information Management) - A NMS (Network Management System) - A CPSIEM (Cyber & Physical Security Information Management System)

### 2.3.5. Operator

The level 1 operator is in charge of receiving manager notifications and identify or confirm when an event should be considered as an incident. The operator must also decide if there is a known resolution for this incident or if it needs a deeper analysys.

### 2.3.6. Analyst

The analyst will be contacted by the operator to analyze complex incidents that can’t be easily resolved. The investigation starts with the IDMEFv2 information but the analyst might need more information like raw logs for a deeper forensics.

### 2.3.7. Attack

An attack is an attempt to destroy, expose, alter, disable, steal or gain unauthorized access to or make unauthorized use of a cyber or physical asset. An attack is one or many kinds of incidents.

### 2.3.8. Correlation

Correlation is the identification of relationships between two or more events.

### 2.3.9. Aggregation

Aggregation is the consolidation of similar events into a single event.

# 3. The IDMEF Data Types

Each object inside the IDMEF data model has an associated data type. This type may be used to validate the content of incoming IDMEF messages.

## 3.1. Classes

The classes are meant to group related attributes together. Some of the classes may be instanciated multiple times (e.g. Source, Target, etc.) while others may only appear once in an IDMEF message (e.g. Analyzer).

## 3.2. Numbers

### 3.2.1. Integers

Integers inside the IDMEF data model are expressed using the following ABNF [RFC5234] grammar:

```
integer         =  *1minus int
int             =  zero / ( digit1-9 *DIGIT )
minus           =  %x2D                          ; -
zero            =  %0x30                         ; 0
digit1-9        =  %x31-39                       ; 1-9
```

E.g. 123.

Such values are indicated with the "INT" type annotation in the model.

### 3.2.2. Floating-point values

Floating-point values inside the IDMEF data model are expressed using the following ABNF grammar:

```
float           =  integer *1frac
frac            =  decimal-point 1*DIGIT
decimal-point   =  %x2E                           ; .
```

This grammar reuses some of the production rules listed in Section 3.2.1.

E.g. 12.34.

Such values are indicated with the "FLOAT" type annotation in the model.

## 3.3. Strings

Strings are series of characters from the [UNICODE] standard and are used to represent a text.

For readability, this document uses quotes (") to delimit strings, but please note that these quotes are not syntactically part of the actual strings.

E.g. "Hello world".

Some of the strings used in the IDMEFv2 data model follow a stricter syntax. These are included below for completeness.

Such values are indicated with the "STRING" type annotation in the model.

### 3.3.1. Enumerations

Enumerations are special strings used when valid values for an IDMEF attribute are restricted to those present in a predefined list.

Such values are indicated with the "ENUM" type annotation in the model.

### 3.3.2. Timestamps

Timestamps are used to indicate a specific moment in time. The timestamps used in the IDMEF data model follow the syntax defined by the "date-time" production rule of the grammar in [RFC3339] ch 5.6.

E.g. "1985-04-12T23:59:59.52Z" represents a moment just before April 5th, 1985 in Coordinated Universal Time (UTC).

Such values are indicated with the "TIMESTAMP" type annotation in the model.

### 3.3.3. Geographical Locations

Some attributes inside the IDMEF data model may refer to geographical locations using a set of coordinates. The reference system for all geographical coordinates is a geographic coordinate reference system, using the World Geodetic System 1984 [WGS84]. The reference system used is the same as for the Global Positioning System (GPS).

The format for such values can be either "latitude,longitude" or "latitude,longitude,altitude". Each of these coordinates is represented as a floating-point value. The latitude and longitude are expressed in degrees while the altitude is expressed in meters.

E.g. "48.8584,2.2945,276.13" matches the (3-dimensional) geographical location for the top floor or the Eiffel Tower located in Paris, France, while "48.8584,2.2945" matches the same location in two dimensions (with the altitude removed).

Such values are indicated with the "GEOLOC" type annotation in the model.

### 3.3.4. UNECE Location Codes (UN/LOCODE)

Some attributes inside the IDMEF data model may refer to geographical locations using Locations Codes. These codes can be assimilated to an enumeration, where the list of possible values is defined in the United Nations Economic Commission for Europe (UNECE) Codes for Trade [UN-LOCODE].

E.g. "FR PAR" is the Location Code for the city of Paris, France.

Such values are indicated with the "UNLOCODE" type annotation in the model.

### 3.3.5. Uniform Resource Identifiers (URIs)

The IDMEF data model uses Uniform Resource Identifiers (URIs), as defined in [RFC3986], when referring to external resources. Unless otherwise specified, either a Uniform Resource Location (URL) or a Uniform Resource Name (URN) may be used where a URI is expected.

E.g. both "https://example.com/resource" and "urn:myapp:resource" are valid Uniform Resource Identifiers.

Such values are indicated with the "URI" type annotation in the model.

### 3.3.6. IP Addresses

IP addresses inside the IDMEF data model are expressed as strings using the traditional dotted-decimal notation for IPv4 addresses (defined by the "dotnum" production rule in the grammar in [RFC5321]), while IPv6 addresses are expressed using the text representation defined in [RFC4291] ch 2.2.

E.g. "192.0.2.1" represents a valid IPv4 address, while "::1/128" represents a valid IPv6 address.

It is RECOMMENDED that implementations follow the recommendations for IPv6 text representation stated in [RFC5952].

Such values are indicated with the "IP" type annotation in the model.

### 3.3.7. E-mail addresses

E-mail addresses inside the IDMEF data model are expressed as strings using the address specification syntax defined in [RFC5322] ch 3.4.1.

E.g. "root@example.com".

Such values are indicated with the "EMAIL" type annotation in the model.

### 3.3.8. Attachment names

Attachments inside the IDMEF data model are identified using a unique name, composed of a string whose character set is limited to the ASCII letters (A-Z a-z) and digits (0-9).

E.g. "state" is a valid name for an attachment.

The constraint on name unicity is enforced per class. That is, but it is not possible for two attachments to share the same name inside the same alert.

Such values are indicated with the "ID" type annotation in the model.

### 3.3.9. Media types

Media types are used in the IDMEF data model to describe an attachment's content. The syntax for such values is defined in [RFC2046].

IANA keeps a list of all currently registered media types in the Media Types registry .

E.g. "application/xml" or "text/plain; charset=utf-8".

Such values are indicated with the "MEDIATYPE" type annotation in the model.

### 3.3.10. Universally Unique IDentifiers (UUIDs)

Universally Unique Identifiers (UUIDs) are used to uniquely identify IDMEF messages. It is also possible for an IDMEF message to reference other IDMEF messages using their UUIDs. The syntax for UUIDs is defined in [RFC4122].

To limit the risk of UUID collisions, implementors SHOULD NOT generate version 4 UUIDs (randomly or pseudo-randomly generated UUIDs).

E.g. "ba2e4ef4-8719-42bb-a712-d6e8871c5c5a".

UUIDs are case-insensitive when used in comparisons.

Such values are indicated with the "UUID" type annotation in the model.

### 3.3.11. Protocol Names

Such values are indicated with the "PROTOCOL" type annotation in the model.

### 3.3.12. IDMEF Paths

This document defines a way to represent the path to every possible attribute inside an IDMEF message. For conciseness, the top-level "Alert" class is omitted from the path.

This representation can be used in contexts where the path to an IDMEF attribute is expected. An example of such usage can be seen in the definition of the "AggrCondition" attribute inside the Alert class (Section 5.2).

The syntax for these IDMEF paths is expressed in the following ABNF grammar:

```
class-name      =  "Analyzer" / "Sensor" / "Source" / "Target" /
                   "Attachment"
attribute-name  =  1*ALPHA
class-reference =  class-name "."
num             =  *1"-" 1*DIGIT
list-index      =  "(" num ")"
path            =  *1class-reference attribute-name *1list-index
```

Valid attribute names are limited to those defined for the specified class-reference (or in the top-level "Alert" class if class-reference is omitted).

For example, the following path refers to the "CeaseTime" attribute of the top-level "Alert" class: "CeaseTime".

Likewise, the following path refers to the "Name" attribute of the "Analyzer" class: "Analyzer.Name".

For attributes defined as lists (see Section 3.4), the path may include the (0-based) index for an entry inside the list. The index defaults to 0 if omitted. This means that several (valid) representations may be used to reference the same IDMEF attribute when list attributes are involved.

For example, both of the following paths refer to the IP address of the first source associated with an IDMEF message:

```
Source.IP
Source(0).IP
```

Compatible implementations MUST reject paths that reference an unknown class, an unknown attribute, or use a list-index for an IDMEF field which is not defined as a list.

A compatible implementation MUST also normalize paths before comparing them (e.g. by stripping the text "(0)" from paths referring to list attributes).

#### 3.3.13. Hashes

Hashes are sometimes used inside the data model to protect the integrity (and optionally, authenticity) of attachments.

The syntax for these values is "function:hash_result", where "function" refers to one of the hashing function names listed in and "hash_result" contains the hexadecimal notation for the hash result obtained by calling the specified hash function on the input value.

In the context of IDMEF, either a keyless or keyed hash function may be used to process the raw input value.

E.g. "sha256:a02735ed8b10ad432d557bd4849c0dac3b23d64706e0618716d6df2def338374"

Hashes are case-insensitive when used in comparisons.

Such values are indicated with the "HASH" type annotation in the model.

### 3.4. Lists

Some attributes of the IDMEF data model accept ordered lists of values.

Such ordered lists are indicated with the "X[]" type annotation in the model. where "X" refers to one of the data types defined in Section 3. For example, "ENUM[]" refers to an ordered list of enumeration values.

# 4. The IDMEF extension

In order to support the dynamic nature of security operations and to adapt to specific needs, the IDMEFv2 data model will need to continue to evolve. This section discusses how new data elements can be incorporated into the IDMEFv2. There is support to add additional enumerated values and new attributes.

These extension mechanisms are designed so that adding new data elements is possible without requiring modifications to this document. Extensions can be implemented publicly or privately. With proven value, well-documented extensions can be incorporated into future versions of the specification.

## 4.1. Extending the Enumerated Values of Attributes

Additional enumerated values can be added to select attributes either through the use of specially marked attributes with the "ext-" prefix or through a set of corresponding IANA registries. The former approach allows for the extension to remain private. The latter approach is public.

### 4.1.1. Private Extension of Enumerated Values

The data model supports adding new enumerated values to an attribute without public registration. For each attribute that supports this extension technique, there is a corresponding attribute in the same element whose name is identical but with a prefix of "ext-". This special attribute is referred to as the extension attribute. The attribute being extended is referred to as an extensible attribute. For example, an extensible attribute named "foo" will have a corresponding extension attribute named "ext-foo". An element may have many extensible attributes.

In addition to a corresponding extension attribute, each extensible attribute has "ext-value" as one its possible enumerated values. Selection of this particular value in an extensible attribute signals that the extension attribute contains data. Otherwise, this "ext-value" value has no meaning.

In order to add a new enumerated value to an extensible attribute, the value of this attribute MUST be set to "ext-value", and the new desired value MUST be set in the corresponding extension attribute. For example, extending the Category attribute of the Analyzer class would look as follows:

```
Analyzer: {
    ...
    "Category":["ext-value"],
    "ext-Category": "my-new-analyzer-category",
    ....
}
```

A given extension attribute MUST NOT be set unless the corresponding extensible attribute has been set to "ext-value".

### 4.1.2. Public Extension of Enumerated Values

The data model also supports publicly extending select enumerated attributes. A new entry can be added by registering a new entry in the appropriate IANA registry. Section (Table 5) provides a mapping between the extensible attributes and their corresponding registry.

## 4.2. Private Extension of Attributes

Use of new attributes is possible through the use of the attachment class. New attributes and their corresponding values should be stored in the Content attribute of an Attachment and the ContentEncoding must be set to JSON. For example creating a new attribute to store the email of the operator (in charge of solving the incident) will look as follows:

```
"Attachment" : [
    {
    "Name": "Operator",
    "ContentEncoding": "JSON",
    "Content": "{\"OperatorMail\":\"John.Does@acme.com\"}",
    }
[
```

# 5. The IDMEF Data Model

In this section, the individual components of the IDMEF data model will be discussed in detail. For each class, the semantics will be described.

## 5.1. Overview

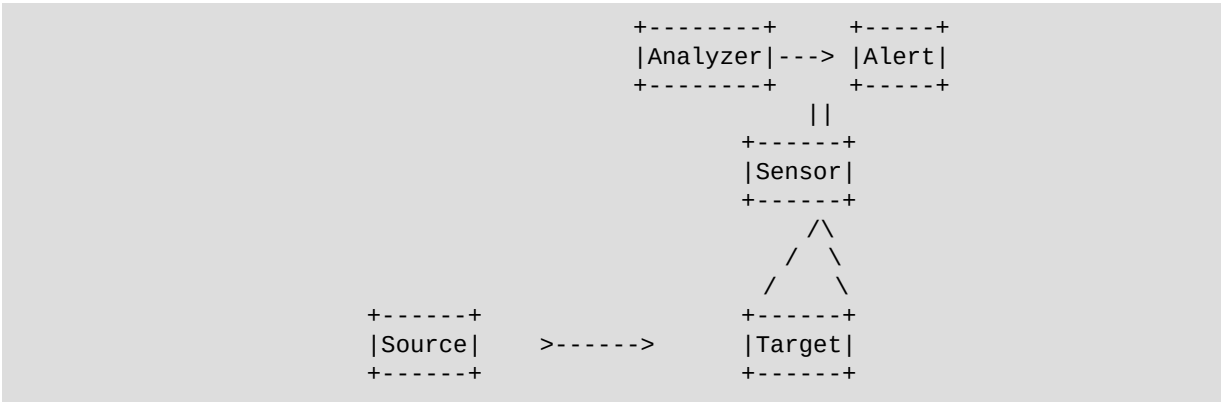



Figure 2: IDMEFv2 Overview Classes

An IDMEF message is composed of an instance of the Alert class (Section 5.2) representing the overall properties of the message. It also contains exactly one instance of the Analyzer class (Section 5.3) and zero or more instances of the Sensor class (Section 5.4). The message may also describe various aspects of an event using the Source (Section 5.5) and Target (Section 5.6) classes.

Last but not least, it may also include zero or more instances of the Attachment class (Section 5.7), e.g. captured files or network packets related to the event for example.

The relationship between the main Alert class and other classes of the data model is shown in Figure 3 (attributes are omitted for clarity).

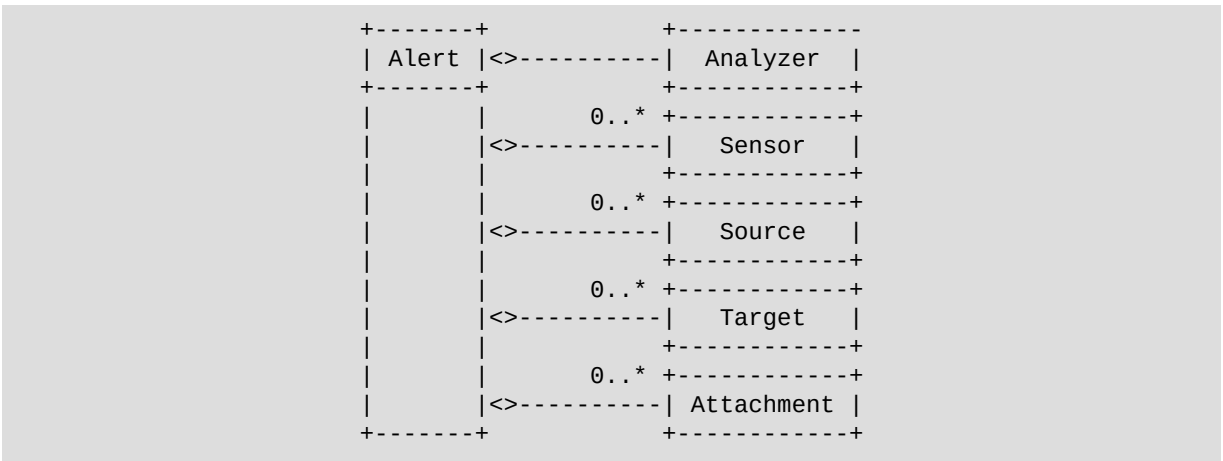



Figure 3: IDMEFv2 Classes

It is important to note that the data model does not specify how an incident should be categorized or identified. For example, an attacker scanning a network for machines listening on a specific port may be identified by one analyzer as a single attack against multiple targets, while another analyzer may identify it as multiple attacks from a single source. However, once an analyzer has determined the type of alert it plans on sending, the data model dictates how that alert should be formatted.

## 5.2. The Alert Class

The Alert class contains high level information about the event that triggered the alert.




Figure 4: The Alert class

The aggregate classes that make up Alert are:

| | |
|---|---|
| Analyzer | Exactly one. An instance of the Analyzer class (Section 5.3) that describes the tool/device responsible for the analysis that resulted in the alert being created and sent. |
| Sensor | Zero or more. Instances of the Sensor class (Section 5.4) used to describe the sensor(s) that captured the information used during the analysis.<br>Depending on the tools/devices used to detect incidents, an Analyzer may rely on the output from a single sensor or from multiple sensors to generate alerts. In addition, the Analyzer and Sensor may actually be part of the same physical device and may share some of their attributes (e.g. IP, Hostname, Model, etc.). |
| Source | Zero or more. Instances of the Source class (Section 5.5) used to describe the source(s) of the incident (e.g. attackers, faulty device, etc.). |
| Target | Zero or more. Instances of the Target class (Section 5.6) used to describe the target(s) of the incident, i.e. the impacted devices/users/services/locations. |

| | |
|---|---|
| Attachment | Zero or more. Instances of the Attachment class (Section 5.7) used to describe the electronic artifacts captured in relation with the event.<br>The intent of the Attachment class is to keep track of the electronic files left as a trail during the event. This may include things like on-disk files (e.g. malware samples), network packet captures, videos or still images from a camera feed, voice recording, etc. |

The Alert class has the following attributes:

| | |
|---|---|
| Version | Mandatory. The version of the IDMEF format in use by this alert.<br>During the drafts tuning period the version is equal to the draft version.<br>Therefore it is "2.D.V0X" for Draft V0X. |
| ID | Mandatory. Unique identifier for the alert. |
| OrganisationName | Optional. Corporate/Main Office Organisation Name<br>Useful if alerts are sent to a multi-organisation central incident detection manager.<br>Example: ACME Corporation |
| OrganisationId | Optional. Corporate/Main Office Organisation ID. Where possible official organisation ID manage by national authority.<br>Useful if alerts are sent to a multi-organisation central incident detection manager.<br>This ID has to be chosen depending on the overall detection perimeter and the nature of the monitored organisation (Private/Public, Commercial, International, etc.)<br>Examples: OrganisationId in France could be SIREN, in England could be CR, Germany could be Handelsregisternummer, Spain could be CIF, Italia could be Partita IVA, USA could be EIN, etc. Commercial OrganisationId in Europe could be V.A.T ID |
| EntityName | Optional. Entity Name, monitored by the organisation, where the incident occurred.<br>Could be a town, region or country name or an internal name. Could also be the name of a customer for a MSSP centralizing it's customer incidents in a single system.<br>Do not repeat the organisation name in the EntityName<br>Example:<br>- ACME HeadQuaters is located in Paris France and has a local office in India<br>- If the incident occurred in the local office: "OrganisationName": "ACME" , "EntityName": "India"<br>- If the incident occurred in the headquaters: "OrganisationName": "ACME", "EntityName": "Headquaters" (or "Paris") |
| EntityId | Optional. Entity ID, monitored by the organisation, where the incident occurred.<br>Useful if organisation and entity are not directly linked, like a customer and a MSSP. |
| EntitySector | Optional. The economic sector(s) and sub-sector(s) in which the entity operates. Values follow the dot notation sector.subsector based on the critical infrastructure taxonomy defined in the NIS2 Directive and CER (Critical Entities Resilience) Directive.<br>This attribute enables sector-based correlation, regulatory compliance reporting, and risk context for incident detection.<br>Enumeration available here Table 6 |

Type — Optional. Incident type.

| Rank | Keyword | Description |
|---|---|---|
| 0 | Cyber | Incident involving cyberspace, information systems, or digital assets. |
| 1 | Physical | Incident involving physical assets, facilities, or infrastructure. |
| 2 | Availability | Incident affecting the availability or performance of a service, system, asset or resource. |
| 3 | Hazards | Incident arising from natural, societal, technological, environmental, or other hazards. |
| 4 | Combined | Incident involving two or more incident types. |

Table 1: Incident types

Category — Optional. Incident category.
Enumeration available here Table 7

ext-Category — Optional. A means by which to extend the Category attribute. (see Section 4.1.1)

Cause — Optional. Incident cause. The cause can be modified by any analyzer on the way of the alert and later by the operator and/or the analyst if new investigation reveals and confirms a different cause of the event.

| Rank | Keyword | Description |
|---|---|---|
| 0 | Normal | The event is related to an expected phenomenon or to a phenomenon that does not qualify as out of the ordinary. |
| 1 | Error | The event is related to a human error. |
| 2 | Malicious | The event is related to malicious code or malicious actions. |
| 3 | Malfunction | The event is related to a device or service malfunction. |
| 4 | Hazard | The event is related to a hazard phenomenon. |
| 5 | Unknown | The cause of the event is unknown. |

Table 2: Incident causes

Description — Optional. Short free text human-readable description of the event.
The description can add detail to the event category for easiest/faster comprehension by the operator. Example : * Cryptoware WannaCry blocked on pegasus server * Unknown person entering through east doorway

Status — Optional. Event state in the overall event lifecycle.

| Rank | Keyword | Description |
|---|---|---|
| 0 | Event | The event is still considered as an harmless event and should not be treated. |
| 1 | Incident | The event is considered as an incident and should be taken care of. |
| 2 | Open | The incident is confirmed and actively being investigated. |
| 3 | Closed | Investigation is finished and the issue is handled. |
| 4 | FalsePositive | Investigation is finished, incident was a false positive. |
| 5 | Obsolete | Incident is obsolete, usually by another incident. |

Table 3: Incident statuses

Priority — Optional. Priority of the alert. Priority is defined by combining impact and urgency. It indicates how fast the incident should be taken care of.

| Rank | Keyword | Description |
|---|---|---|
| 0 | Unknown | Priority unknown |
| 1 | Info | Informational event with no current impact; no action or follow-up is required. |
| 2 | Low | Minor incident with negligible impact; a workaround exists, and resolution can be scheduled. |
| 3 | Medium | Moderate incident affecting a limited scope; requires timely attention but is not immediately business-threatening. |
| 4 | High | Significant incident with major service degradation or impact on important assets; demands an urgent response. |
| 5 | Critical | Severe incident with business-critical systems down or vital interests threatened; requires immediate action and escalation. |

Table 4: Incident priorities

Confidence — Optional. A floating-point value between 0 and 1 indicating the analyzer's confidence in its own reliability of this particular detection, where 0 means that the detection is surely incorrect while 1 means there is no doubt about the detection made.

Note — Optional. Free text human-readable additional note, possibly a longer description of the incident if is not already obvious.
The Note attribute can be used to store any additional information. It can be additional information about the event and/or about the incident resolution,

| | |
|---|---|
| | although the incident resolution information should in principle be stored elsewhere (with a link with the external tool in AltNames) |
| CreateTime | Mandatory. Timestamp indicating when the alert was created. |
| StartTime | Optional. Timestamp indicating the deduced start of the event.<br>StartTime can be later than CreateTime in case or Alerts created from forecast information (e.g. Snow Storm expected in two days staring at 10h00) |
| EndTime | Optional. Timestamp indicating the deduced end of the event. |
| ReportTime | Optional. Timestamp indicating the reporting time of the event, usually to an external CSIRT or a central SIEM. |
| AltNames | Optional. Alternative identifiers; strings which help pair the event to internal systems' information (for example ticket IDs inside a request tracking systems). |
| AltCategory | Optional. Alternate categories from a reference other than IDMEFv2 categories (e.g. MISP, MITRE ATT@CK or another proprietary/internal reference). |
| Ref | Optional. References to sources of information related to the incident and/or vulnerability, and specific to this incident.<br>This MAY be a URL to additional info, or a URN in a registered or unregistered ad-hoc namespace bearing reasonable information value and uniqueness, such as "urn:cve:CVE-2013-2266". |
| CorrelID | Optional. Identifiers for the messages which were used as information sources to create this message, in case the message has been created based on correlation/analysis/deduction from other messages. |
| AggrCondition | Optional. A list of IDMEF fields used to aggregate events. The values for these fields will be the same in all aggregated events.<br>This attribute should mostly be set by intermediary nodes, which detect duplicates, or aggregate events, spanning multiple detection windows, into a longer one.<br>The "StartTime" and "EndTime" attributes are used in conjunction with this attribute to describe the aggregation window. |
| PredID | Optional. A list containing the identifiers of previous messages which are obsoleted by this message.<br>The obsoleted alerts SHOULD NOT be used anymore. This field can be used to "update" an alert. |
| RelID | Optional. A list containing the identifiers of other messages related to this message. |

## 5.3. The Analyzer Class

The Analyzer class describes the module that has analyzed the data captured by the sensors, identified an event of interest and decided to create an alert.

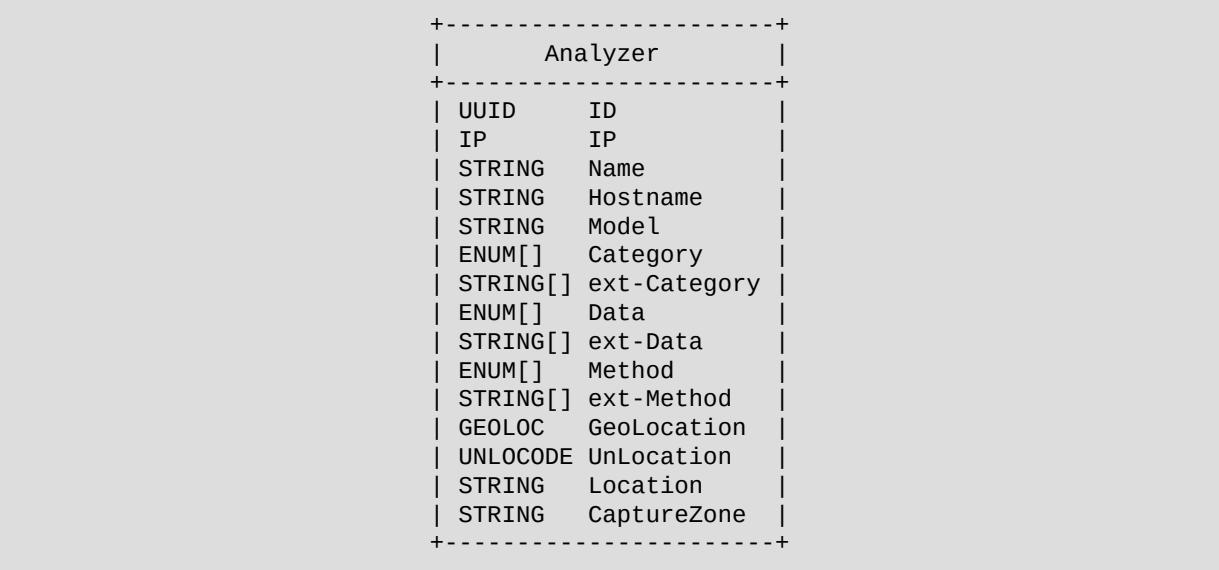



Figure 5: The Analyzer class

The Analyzer class has the following attributes:

| | |
|---|---|
| ID | Optional. Unique identifier for the analyzer. |
| IP | Optional. Analyzer IP address. |
| Name | Recommended. Name of the analyzer, which must be reasonably unique, however still bear some meaningful sense.<br>This attribute usually denotes the hierarchy of organizational units the detector belongs to and its own name. It MAY also be used to distinguish multiple analyzers running with the same IP address. |
| Hostname | Optional. Hostname of this analyzer.<br>SHOULD be a fully-qualified domain name. |
| Model | Optional. Analyzer model description (usually its generic name, brand and version). |
| Category | Optional. Analyzer categories.<br>Enumeration available here Table 8 |
| ext-Category | Optional. A means by which to extend the Category attribute. (see Section 4.1.1) |
| Data | Optional. Type of data analyzed during the detection.<br>Enumeration available here Table 9 |
| ext-Data | Optional. A means by which to extend the Data attribute. (see Section 4.1.1) |
| Method | Optional. Detection method.<br>Enumeration available here Table 10 |
| ext-Method | Optional. A means by which to extend the Method attribute. (see Section 4.1.1) |
| GeoLocation | Optional. GPS coordinates for the analyzer. |
| UnLocation | Optional. Standard UN/Locode for the analyzer. |
| Location | Optional. Internal name for the location of the analyzer. |

CaptureZone — Optional. A string that describes the "capture zone" of the analyzer (if the analyzer is also a sensor), as a JSON-serialized string.
Depending on the type of sensor, the capture zone may for instance refer to:
A JSON object describing a camera's settings (elevation, horizontal and vertical field of view, azimuth, etc.)
A description of the IP network where packet capture is taking place.

## 5.4. The Sensor Class

The Sensor class describes the module that captured the data before sending it to an analyzer. The Sensor may be a subpart of the Analyzer.

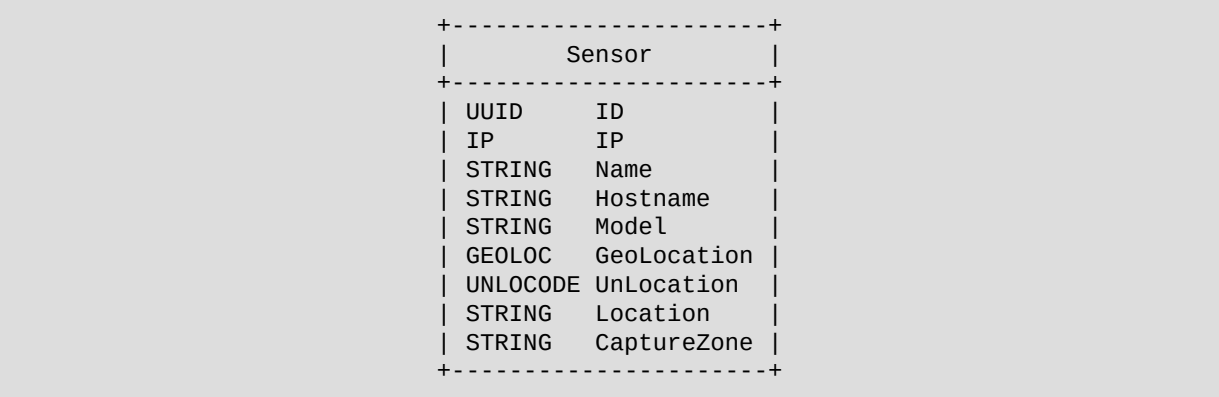



Figure 6: The Sensor class

The Sensor class has the following attributes:

ID — Optional. Unique identifier for the sensor.

IP — Optional. The sensor's IP address.

Name — Recommended. Name of the sensor, which must be reasonably unique, however still bear some meaningful sense.
This attribute usually denotes the hierarchy of organizational units the sensor belongs to and its own name. It MAY also be used to distinguish multiple sensors running with the same IP address.

Hostname — Optional. The sensor's hostname.
This SHOULD be a fully qualified domain name, but may not conform exactly because values extracted from logs, messages, DNS, etc. may themselves be malformed.
An empty string MAY be used to explicitly state that this value was inquired but not found (missing DNS entry).

Model — Optional. The sensor model's description (usually its generic name, brand and version).

GeoLocation — Optional. GPS coordinates for the analyzer.

UnLocation — Optional. Standard UN/Locode for the sensor.

Location — Optional. Internal name for the location of the sensor.

CaptureZone — Optional. A string that describes the "capture zone" of the sensor, as a JSON-serialized string.
Depending on the type of sensor, the capture zone may for instance refer to:

- A JSON object describing a camera's settings (elevation, horizontal and vertical field of view, azimuth, etc.)
- A description of the IP network where packet capture is taking place.

## 5.5. The Source Class

The Source class describes the origin(s) of the event(s) leading up to the creation of this alert.

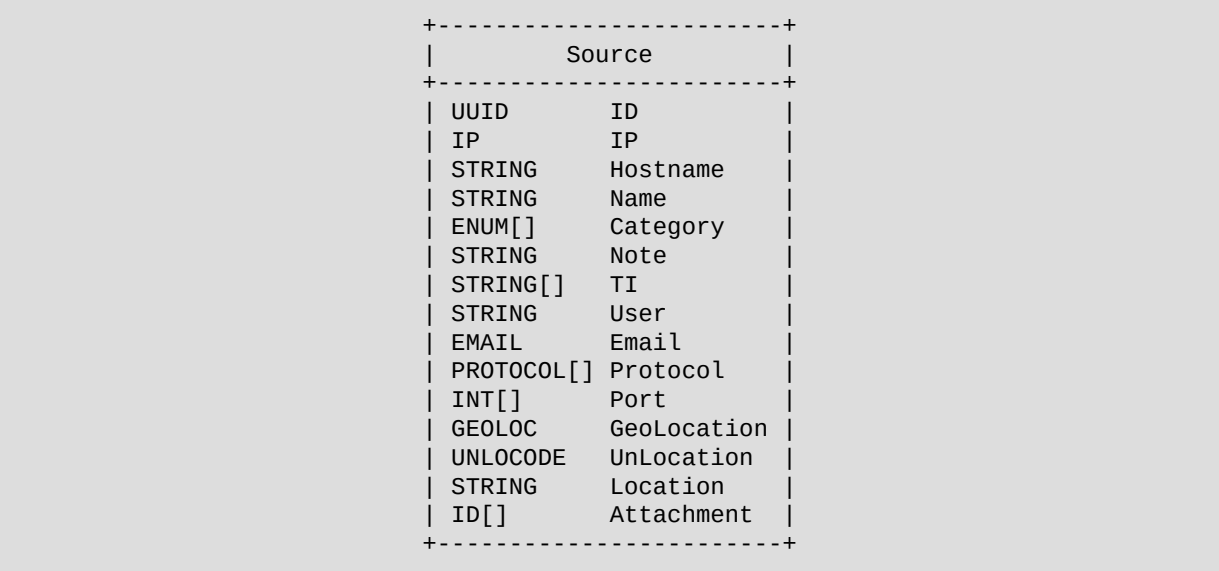



Figure 7: The Source class

The Source class has the following attributes:

| | |
|---|---|
| ID | Optional. Unique identifier for the source. |
| IP | Optional. Source IP address. |
| Hostname | Optional. Hostname of this source.<br>This SHOULD be a fully qualified domain name, but may not conform exactly because values extracted from logs, messages, DNS, etc. may themselves be malformed.<br>An empty string MAY be used to explicitly state that this value was inquired but not found (missing DNS entry). |
| Name | Optional. Name of this source.<br>Naming the source might be necessary for non digital source like a human, a vehicle, a drone, etc. |
| Category | Optional. Source category<br>Enumeration available here Table 11 |
| Note | Optional. Free text human-readable additional note for this source. |
| TI | Optional. Threat Intelligence data about the source.<br>Values in this list MUST use the format "attribute:origin", where "attribute" refers to the attribute inside this source found inside a Threat Intelligence database, and "origin" contains a short identifier for the Threat Intelligence database. E.g. "IP:Dshield".<br>Please note that the same attribute may appear multiple times inside the list (because a match was found in multiple Threat Intelligence databases). |
| User | Optional. User ID or login responsible for the incident. |
| Email | Optional. Email address responsible for the incident.<br>E.g. the value of the "Reply-To" or "From" header inside a phishing e-mail. |
| Protocol | Optional. Protocols related to connections from/to this source.<br>If several protocols are stacked, they MUST be ordered from the lowest (the closest to the medium) to the highest (the closest to the application) according to the ISO/OSI model. |

| | |
|---|---|
| Port | Optional. Source ports involved in the incident.<br>Values in this list MUST be integers and MUST be in the range 1-65535. |
| GeoLocation | Optional. GPS coordinates for the source. |
| UnLocation | Optional. Standard UN/Locode for the source. |
| Location | Optional. Internal name for the location of the source. |
| Attachment | Optional. Identifiers for attachments related to this source.<br>Each identifier listed here MUST match the "Name" attribute for one of the attachments described using the Attachment class (Section 5.7). |

## 5.6. The Target Class

The Target class describes the target(s) impacted by the event(s) leading up to the creation of this alert.

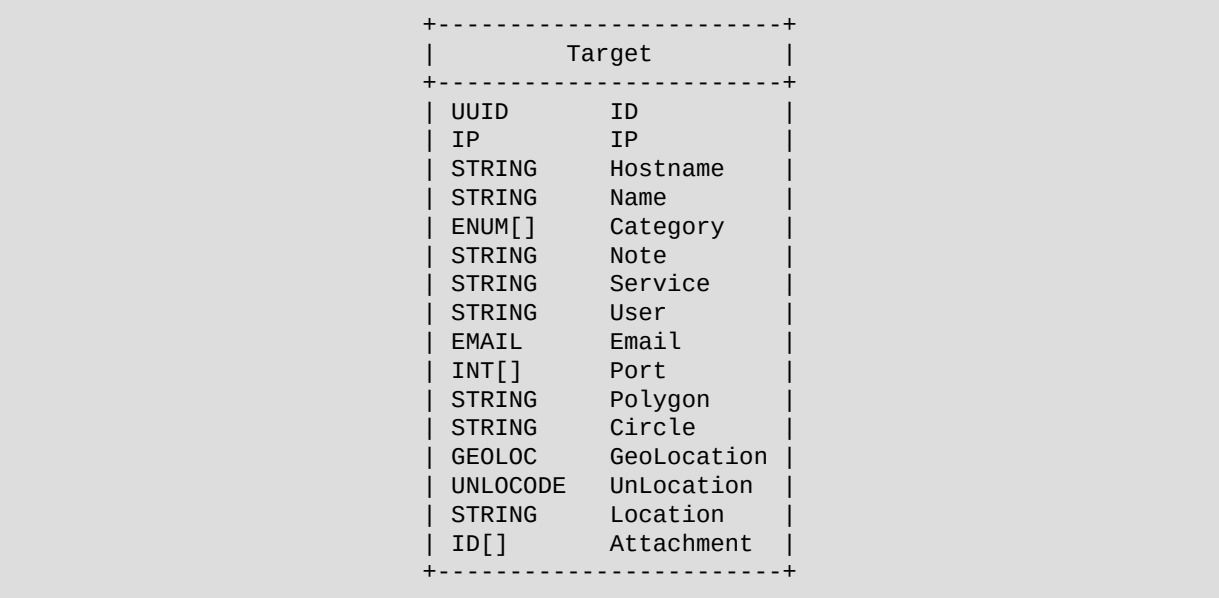



Figure 8: The Target class

The Target class has the following attributes:

| | |
|---|---|
| ID | Optional.Unique identifier for the target. |
| IP | Optional. Target IP address. |
| Hostname | Optional. Hostname of this target.<br>This SHOULD be a fully qualified domain name, but may not conform exactly because values extracted from logs, messages, DNS, etc. may themselves be malformed.<br>An empty string MAY be used to explicitly state that this value was inquired but not found (missing DNS entry). |
| Name | Optional. Name of this target.<br>Naming the target might be necessary for non digital targets like a human, a vehicle, a building, etc. |
| Category | Optional. Target category<br>Enumeration available here Table 12 |
| Note | Optional. Free text human-readable additional note for this target. |
| Service | Optional. Should be High Level Service (messaging, internet, ...), if not defined can be process impacted by the incident. |

| | |
|---|---|
| User | Optional. User ID or login impacted by the incident. |
| Email | Optional. Email address impacted by the incident.<br>E.g. the value of the "To" header inside a phishing e-mail. |
| Port | Optional. Target ports involved in the incident.<br>Values in this list MUST be integers and MUST be in the range 1-65535. |
| Polygon | Optional. Pollygonal incident affected area. Whitespace-delimited list of WGS 84 'latitude,longitude' pairs. Minimum 4 pairs, first equals last |
| Circle | Optional. Circular incident infected area. WGS 84 center point 'latitude,longitude' followed by a space and radius in kilometers |
| GeoLocation | Optional. GPS coordinates for the target. |
| UnLocation | Optional. Standard UN/Locode for the target. |
| Location | Optional. Internal name for the location of the target. |
| Attachment | Optional. Identifiers for attachments related to this target.<br>Each identifier listed here MUST match the "Name" attribute for one of the attachments described using the Attachment class (Section 5.7). |

## 5.7. The Attachment Class

The Attachment class contains additional data which was captured in relation with the event.

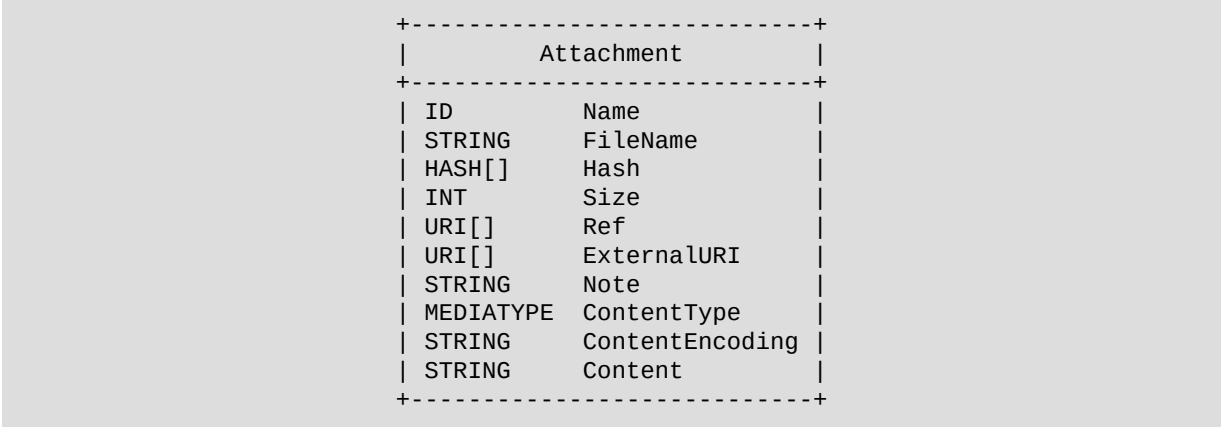



Figure 9: The Attachment class

The Attachment class has the following attributes:

| | |
|---|---|
| Name | Mandatory. A unique identifier among attachments that can be used to reference this attachment from other classes using the "Attachment" attribute. |
| FileName | Optional. Attachment filename.<br>This will usually be the original name of the captured file or the name of the file containing the captured content (e.g. a packet capture file). |
| Hash | Optional. A list of hash results for the attachment's Content.<br>The values in this list are computed by taking the raw value of the attachment's "Content" attribute. The hash result is computed before any other transformation (e.g. Base64 encoding) is applied to the content, so that a receiving IDMEF system may reverse the transformation, apply the same hashing function and |

| | |
|---|---|
| | obtain the same hash result. See also the definition for the "ContentEncoding" attribute below.<br>It is RECOMMENDED that compatible implementations use one of the hashing functions from the SHA-2 [RFC6234] or SHA-3 [NIST.FIPS.202] families to compute the hash results in this list. |
| Size | Optional. Length of the content (in bytes).<br>This value MUST be a non-negative integer. |
| Ref | Optional. References to sources of information related to the incident and/or vulnerability, and specific to this attachment. |
| ExternalURI | Optional. If the attachment's content is available and/or recognizable from an external resource, this is the URI (usually a URL) to that resource.<br>This MAY also be a URN in a registered or unregistered ad-hoc namespace bearing reasonable information value and uniqueness, such as "urn:mhr:55eaf7effadc07f866d1eaed9c64e7ee49fe081a" or "magnet:?xt=urn:sha1:YNCKHTQCWBTRNJIV4WNAE52SJUQCZO5C". |
| Note | Optional. Free text human-readable additional note for this attachment. |
| ContentType | Optional. Internet Media Type of the attachment.<br>For compatibility reasons, implementations SHOULD prefer one of the well-known media types registered in IANA . |
| ContentEncoding | Optional. Content encoding.<br>The following encodings are defined in this version of the specification:<br>• "json": The content refers to a JSON object which has been serialized to a string using the serialization procedure defined in [RFC8259].<br>• "base64": The content has been serialized using the Base64 encoding defined in [RFC4648].<br>The "base64" encoding SHOULD be used when the content contains binary data.<br>If omitted, the "json" encoding MUST be assumed. |
| Content | Optional. The attachment's content, in case it is directly embedded inside the message.<br>For large attachments, it is RECOMMENDED that implementations make use of the "ExternalURI" attribute to refererence a copy of the content saved in an external storage mechanism. |

## 5.8. The JavaScript Object Notation Serialization Method

This serialization method aims to convert IDMEFv2 messages to a format that is easy to parse and process, both by software/hardware processors, as well as humans. It relies on the the JavaScript Object Notation (JSON) Data Interchange Format defined in [RFC8259].

Conforming implementations MUST implement all the requirements specified in [RFC8259].

In addition, the following rules MUST be observed when serializing an IDMEFv2 message:

- The top-level Alert class (Section 5.2) is represented as a JSON object ([RFC8259]). This JSON object is returned to the calling process at the end of the serialization process.
- Aggregate classes are represented as JSON objects and stored as members of the top-level JSON object, using the same name as in the IDMEF data model. E.g. the appears under the name "Analyzer" inside the top-level JSON object.
- Attributes are stored as members of the JSON object representing the class they belong to, using the same name as in the IDMEF data model. E.g. the "Version" attribute from the is stored under the name "Version" inside the top-level JSON object.

- Lists from the IDMEF data model are represented as JSON arrays ([RFC8259]). This also applies to aggregate classes where a list is expected. E.g. the "Sensor" member inside the top-level JSON object contains a list of objects, where each object represents an instance of the .
- The various string-based data types listed in Section 3 are represented as JSON strings ([RFC8259]). Please note that the issues outlined in [RFC8259] regarding strings processing also apply here.
- IDMEF attributes with the "NUMBER" data type are represented as JSON numbers ([RFC8259]).

# 6. Security Considerations

This document describes a data representation for exchanging security-related information between incident detection system implementations. Although there are no security concerns directly applicable to the format of this data, the data itself may contain security-sensitive information whose confidentiality, integrity, and/or availability may need to be protected.

This suggests that the systems used to collect, transmit, process, and store this data should be protected against unauthorized use and that the data itself should be protected against unauthorized access.

The underlying messaging format and protocol used to exchange instances of the IDMEF MUST provide appropriate guarantees of confidentiality, integrity, and authenticity. The use of a standardized security protocol is encouraged.

The draft-lehmann-idmefv2-https-transport-01.txt document defines the transportation of IDMEF over HTTPs that provides such security.

# 7. IANA Considerations

This document creates 10 identically structured registries to be managed by IANA:

- Name of the registry group: "Incident Detection Message Exchange Format v2 (IDMEF)"
- URL of the registry: http://www.iana.org/assignments/idmefv2
- Namespace format: A registry entry consists of:
  - Rank. A uniq integer for this namespace. Range starts at 0 and ends at the length of this list. The maximum length of this list is 255.
  - Keyword. A keyword for a given IDMEF attribute. It MUST conform to the formatting specified by the IDMEF "ENUM" data type (Section 3.3.1).
  - Description. A short description of the enumerated keyword.
  - Reference. An optional list of URIs to further describe the value.
- Allocation policy: Expert Review per [RFC8126]. This reviewer will ensure that the requested registry entry conforms to the prescribed formatting. The reviewer will also ensure that the entry is an appropriate value for the attribute per the information model (Section 5).

The registries to be created are named in the "Registry Name" column of Table 5. Each registry is initially populated with ranks, keywords and descriptions that come from an attribute specified in the IDMEF model (Section 5). The initial Ranks, Keywords and Description fields of a given registry are listed in "Initial Values". The "Initial Values" column points to a table in this document that lists and describes each enumerated keyword. Each enumerated keyword in the table gets a corresponding entry in a given registry. The initial value of the Reference field of every registry entry described below should be this document.

| **Registry Name** | **Initial Values** |
|---|---|
| Alert-Type | Table 1 (Alert class (Section 5.2)) |
| Alert-Category | (Alert class (Section 5.2)) |
| Alert-Cause | Table 2 (Alert class (Section 5.2)) |
| Alert-Priority | Table 4 (Alert class (Section 5.2)) |
| Alert-Status | Table 3 (Alert class (Section 5.2)) |
| Analyzer-Category | (Alert class (Section 5.2)) |
| Analyzer-Data | (Analyzer class (Section 5.3)) |
| Analyzer-Method | (Analyzer class (Section 5.3)) |

Table 5: IANA Enumerated Value Registries

# 8. Acknowledgement

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- The H2020 7SHIELD project (Safety and Security Standards of Space Systems, ground Segments and Satellite data assets , via prevention, detection, response and mitigation of physical and cyber threats) who implemented in real scale first versions of IDMEFv2 on five pilots around Europe helping greatly to improve it.
- The CESNET team for their work on the [IDEA0] format (based on IDMEFv1) which inspired multiples concepts to IDMEFv2.

# Using IDMEFv2 in Ephemeral Environments (Informative) #
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# Appendix A. Using IDMEFv2 in Ephemeral Environments

## A.1. The Ephemeral Asset Problem

Modern container orchestration platforms and cloud runtime environments create and terminate resources dynamically. IP addresses may change or be reassigned, resource names may be reused, and an entity involved in a security incident may no longer exist when an investigation takes place.

Consequently, identifying a resource solely by its IP address or hostname is insufficient for reliable incident correlation and post-incident investigation.

IDMEFv2 provides complementary attributes that can describe ephemeral entities in the , , , and classes. The attribute identifies an entity in the IDMEFv2 data model, while , , and provide additional information about its network presence, operational hostname, and infrastructure context.

These attributes serve different purposes and SHOULD be used together when the relevant information is available. They do not, by themselves, guarantee the long-term availability of historical records or the continued existence of the resource.

## A.2. Complementary Identification Attributes

The following attributes have complementary roles when describing ephemeral resources.

### A.2.1. ID

The attribute identifies an entity in the IDMEFv2 data model. It MUST be populated in accordance with the applicable IDMEFv2 specification.

An IDMEFv2 identifier and a platform-native identifier are different concepts. For example, a Kubernetes pod UID identifies a Kubernetes object instance; it is not a substitute for the IDMEFv2 attribute. Implementations that need to correlate multiple alerts concerning the same resource SHOULD define a consistent mapping between the resource identity and the IDMEFv2 entity identifier. They SHOULD also distinguish successive resource instances when a platform reuses an operational name.

The identifier MUST NOT be assumed to provide access to a resource or to its historical metadata after termination.

### A.2.2. IP

The attribute records the network address associated with the entity in the context of the event.

It supports correlation with network telemetry, firewall logs, flow records, and other security monitoring data. However, an IP address is a time-dependent network attribute and MUST NOT be treated as a persistent identifier of an ephemeral resource.

When available, the event timestamp and relevant network context SHOULD be retained to support the interpretation of the address at the time of the incident.

Implementations SHOULD distinguish addresses belonging to different infrastructure layers, such as a pod, container, node, NAT gateway, or proxy, when this distinction is relevant to the investigation.

### A.2.3. Hostname

The attribute records the hostname observed or configured for the entity. It SHOULD contain an actual hostname, not an unrelated platform identifier.

For a Kubernetes pod, the hostname can be configured through the pod's hostname settings or obtained from the runtime environment. A pod name and the hostname used inside the pod may happen to be the same, but they are not interchangeable by definition. Implementations SHOULD record the actual hostname observed or configured for the pod.

Hostnames can facilitate operational identification and correlation during a resource's lifetime. However, hostnames may be reused, and their uniqueness may be limited to a particular namespace, cluster, runtime, or time interval. Consequently, SHOULD NOT be considered a globally unique or permanently stable identifier without additional context.

### A.2.4. Location

The attribute can carry a platform-specific resource reference that provides additional context for identifying an entity within its infrastructure.

Depending on the environment, this reference MAY contain a cloud resource identifier, an orchestration resource path, a container identifier qualified by its runtime, or a combination of identifiers that uniquely identifies the relevant resource within its scope.

Implementations SHOULD prefer references that identify the individual resource involved in the event rather than only its parent cluster, service, or workload definition.

Where appropriate, a reference MAY include additional context, such as a cluster identifier, namespace, node identifier, container identifier, or resource UID.

Implementations using platform-specific references SHOULD document their syntax, scope, uniqueness properties, and interpretation.

A value does not, by itself, guarantee that a terminated resource can be retrieved or that its historical metadata remains available. Systems requiring post-incident traceability MUST retain the relevant historical metadata and security telemetry for the required investigation period.

## A.3. Platform-Specific Resource References

The examples below illustrate how can be populated in different environments. The URI schemes used in these examples, including , , , , and , are illustrative implementation conventions. They are not defined or standardized by IDMEFv2.

The values are illustrative and MUST be replaced with identifiers and hostnames obtained from the relevant platform in an actual deployment. UUID-shaped values shown in the examples are illustrative IDMEFv2 entity identifiers; they are not Kubernetes UIDs unless explicitly placed in a platform reference.

### A.3.1. Kubernetes

In Kubernetes, a pod name alone may not be sufficient to distinguish successive instances that share the same name. The pod UID SHOULD be retained when available.

A reference SHOULD include the cluster context, namespace, pod name, and pod UID. Container-specific information MAY be added when the investigation concerns an individual container.

The values below identify IDMEFv2 entities. The values are illustrative actual hostnames configured or observed for the respective pods; they are not merely Kubernetes pod names. The pod names and UIDs are retained in .

Example:

```
{
  "CreateTime": "2026-10-19T10:15:30Z",
  "Analyzer": [
    {
      "ID": "7d6a3c21-1b4e-4f80-9a12-6e5b2c8d901f",
      "Hostname": "secops-collector-01.example.net",
      "Location": "k8s://cluster-secops/namespace-monitoring/pod/idmefv2-
analyzer?uid=11111111-1111-4111-8111-111111111111"
    }
  ],
  "Sensor": [
    {
      "ID": "8e7b4d32-2c5f-4a91-8b23-7f6c3d9e012a",
      "IP": "10.10.0.12",
      "Hostname": "network-sensor-01.example.net",
      "Location": "k8s://cluster-prod-eu/namespace-monitoring/pod/network-
sensor?uid=22222222-2222-4222-8222-222222222222"
    }
  ],
  "Source": [
    {
      "ID": "9f8c5e43-3d60-4ba2-9c34-8a7d4e0f123b",
      "IP": "10.244.2.15",
      "Hostname": "payment-worker-01.namespace-paiement.example.net",
      "Location": "k8s://cluster-prod-eu/namespace-paiement/
pod/payment-worker-5c8d9f-abc123?uid=6a1f8b42-3c9d-4e70-
a521-8b7d9c2e104f&container=worker&node=node-worker-03"
    }
  ],
  "Target": [
    {
      "ID": "a09d6f54-4e71-4cb3-8d45-9b8e5f10234c",
      "IP": "10.244.3.20",
      "Hostname": "payment-api-01.namespace-paiement.example.net",
      "Location": "k8s://cluster-prod-eu/namespace-paiement/pod/payment-
api-7b8c9d-xyz789?
uid=33333333-3333-4333-8333-333333333333&container=api&node=node-
worker-04",
      "Service": "finance"
    }
  ]
}
```

The reference is an illustrative convention. The pod UID identifies the Kubernetes object instance, while the cluster and namespace provide the context required to interpret the reference.

Node-level information MAY be included when it helps correlate the event with network sensors or host-level telemetry.

### A.3.2. Docker and Podman

For standalone container runtimes such as Docker and Podman, a container name alone may be insufficient because names can be reused after a container is removed.

SHOULD identify the container instance and, where necessary, the runtime host or engine. The full container identifier SHOULD be preferred over the human-readable container name.

Example:

```
{
  "Target": [
    {
      "ID": "b10e7a65-5f82-4dc4-9e56-0c9f6a21345d",
      "IP": "172.17.0.3",
      "Hostname": "web-app-01.example.net",
      "Location": "docker://server-web-01/containers/
a1b2c3d4e5f67890123456789012345678901234567890123456789012345678",
      "Service": "e-commerce"
    }
  ]
}
```

The scheme is an illustrative convention, not an official Docker URI scheme. A corresponding convention MAY be defined for Podman, for example using a scheme. Implementations SHOULD document whether the reference identifies the container, the runtime instance, the host, or a combination of these entities.

Where multiple runtime hosts or engines are involved, the reference SHOULD include enough context to avoid ambiguity.

### A.3.3. Google Cloud GKE

For Google Kubernetes Engine (GKE), a reference can combine the Google Cloud project and cluster context with the Kubernetes namespace, pod name, and pod UID.

The cluster resource identifier alone identifies the cluster, not the individual pod involved in the incident.

Example:

```
{
  "Target": [
    {
      "ID": "c21f8b76-6083-4ed5-af67-1d0a7b32456e",
      "IP": "10.244.2.15",
      "Hostname": "payment-worker-01.namespace-paiement.example.net",
      "Location": "gke://projects/my-project-123/locations/us-
west1/clusters/cluster-prod-eu/namespaces/namespace-paiement/
pods/payment-worker-5c8d9f-abc123?uid=6a1f8b42-3c9d-4e70-
a521-8b7d9c2e104f&container=worker",
      "Service": "finance"
    }
  ]
}
```

The scheme is an illustrative convention rather than an official Google Cloud resource URI format. Implementations MAY instead use a documented Google Cloud resource identifier together with the Kubernetes identifiers required to identify the individual pod.

### A.3.4. AWS ECS

In Amazon Elastic Container Service (ECS), the task ARN identifies a task instance, whereas the task definition ARN identifies a versioned task definition.

For incident investigation involving a particular running or terminated task, SHOULD identify the task instance rather than only its task definition. The task definition ARN MAY be retained separately as supplementary configuration context.

Example:

```
{
  "Target": [
    {
      "ID": "d32f8a76-4b19-4c52-9d63-0a8e7f6b1234",
      "IP": "10.0.1.42",
      "Hostname": "payment-worker-01.example.net",
      "Location": "arn:aws:ecs:us-west-1:123456789012:task/cluster-prod-
eu/0123456789abcdef0123456789abcdef",
      "Service": "finance"
    }
  ]
}
```

The ARN is illustrative. Implementations MUST use the actual task ARN returned by ECS and SHOULD retain the cluster context when needed to interpret the reference.

When the incident concerns an individual container within a task, the relevant container identifier SHOULD also be retained when available.

The task definition ARN MAY be recorded as additional information to identify the configuration used by the task. It MUST NOT be substituted for the task instance identifier when the objective is to distinguish individual task executions.

### A.3.5. Microsoft Azure AKS

In Azure Kubernetes Service (AKS), the Azure Resource Manager (ARM) resource ID of a managed cluster identifies the cluster, not an individual pod.

A pod-level reference SHOULD therefore combine the Azure subscription, resource group, and cluster context with the Kubernetes namespace, pod name, and pod UID.

Example:

```
{
  "Target": [
    {
      "ID": "e43f9b87-5c20-4d63-8e74-1b9f6a20345d",
      "IP": "10.240.0.15",
      "Hostname": "payment-worker-01.namespace-paiement.example.net",
      "Location": "aks://
subscriptions/12345678-1234-1234-1234-123456789012/resourceGroups/my-
rg/managedClusters/cluster-prod-eu/namespaces/namespace-paiement/pods/
payment-worker-5c8d9f-abc123?uid=6a1f8b42-3c9d-4e70-a521-8b7d9c2e104f",
      "Service": "finance"
    }
  ]
}
```

The scheme is an illustrative convention, not an official ARM resource identifier for a pod.

Implementations MAY retain the cluster ARM resource ID as a separate reference or incorporate it into a documented platform-specific representation. The reference SHOULD distinguish the individual Kubernetes resource from its parent cluster.

### A.3.6. HashiCorp Nomad

In HashiCorp Nomad, an allocation identifier can provide a reference to a particular workload allocation. The reference SHOULD include sufficient context to distinguish allocations across clusters, regions, namespaces, or jobs, as applicable.

Example:

```
{
  "Target": [
    {
      "ID": "f54a0c98-6d31-4e75-9a82-2c7f1b60345e",
      "IP": "172.17.0.5",
      "Hostname": "payment-worker-01.example.net",
      "Location": "nomad://region-eu/cluster-prod/job-payment/
alloc-01234567-89ab-cdef-0123-456789abcdef?task=worker",
      "Service": "finance"
    }
  ]
}
```

The scheme is an illustrative convention. Implementations SHOULD use the actual allocation identifier provided by Nomad and document the context required to interpret it.

Task-level or container-level information MAY be included when necessary to distinguish the specific runtime entity involved in the event.

## A.4. Scope of the Service Attribute

The attribute in the class SHOULD describe the high-level service or process affected, consistently with the IDMEFv2 specification.

For example, values such as , , and can identify business-level services associated with the target.

Implementations SHOULD avoid using as an undocumented substitute for platform-specific resource identifiers, such as pod names, container IDs, allocation IDs, task ARNs, or cluster resource IDs.

Where business services and technical runtime services must be distinguished, implementations SHOULD document the mapping between these concepts and the relevant IDMEFv2 attributes.

The use of for a particular purpose MUST remain consistent with the attribute definition in the applicable version of the IDMEFv2 specification.

## A.5. Temporal Context and Historical Retention

Reliable post-incident investigation requires more than a stable resource reference.

Implementations SHOULD preserve the event timestamp and sufficient contextual information to determine which resource instance, network address, and runtime environment were involved at the time of the event.

The IDMEFv2 time attributes SHOULD be used according to their defined semantics. In particular, indicates when the alert was created, whereas describes the start of the activity or situation being reported when that information is available. They are not interchangeable, and neither should be assumed to represent the exact measurement time of every individual attribute. Implementations SHOULD retain any additional timestamps required to interpret time-sensitive measurements accurately.

Systems requiring long-term traceability MUST retain the historical metadata and security telemetry needed for the intended investigation period. Such information can include orchestration events, container lifecycle records, cloud audit logs, network flow records, and mappings between resource identifiers and network addresses.

A value MAY be used as a lookup key into retained historical records. However, the availability and retention of those records are properties of the surrounding monitoring and logging infrastructure, not guarantees provided by the attribute itself.

## A.6. Summary

Ephemeral environments require a distinction between entity identification, network correlation, and historical resource context.

The attribute identifies the entity in the IDMEFv2 data model. records the network address observed in the context of the event, provides the actual configured or observed hostname, and can carry a platform-specific reference to the relevant resource.

Implementations SHOULD use resource references that identify individual instances rather than only their parent clusters or configuration definitions. They SHOULD preserve sufficient temporal and infrastructure context to distinguish successive instances and correlate events across security telemetry sources.

Long-term traceability additionally depends on the retention and availability of historical metadata and security records.

This approach enables IDMEFv2 to represent ephemeral resources across different orchestration platforms and container runtimes without requiring a separate IDMEFv2 extension for each environment. Platform-specific reference formats remain implementation conventions and SHOULD be documented to support interoperability.

The JSON fragments are illustrative. Their exact structure MUST conform to the applicable IDMEFv2 serialization and schema.

# Appendix B. Enumerations

## B.1. Entity Sector

| Rank | Keyword | Definition |
|---|---|---|
| 0 | Undefined | Sector undefined |
| 1 | Administration.CentralGovernment | Central government ministries and agencies |
| 2 | Administration.Diplomatic | Embassies, consulates, and diplomatic missions |
| 3 | Administration.EmergencyServices | Police, fire, rescue, and emergency medical services |
| 4 | Administration.Judiciary | Courts, judicial systems, and correctional facilities |
| 5 | Administration.Local | Regional and municipal public services |
| 6 | Administration.Other | Other public administration entities critical for the continuity of public services |
| 7 | Banking.Banking | Banking institutions and credit activities |
| 8 | Banking.Other | Other banking and financial entities critical for the stability of the financial system |
| 9 | Cemeteries.Crematoria | Crematoria operations |
| 10 | Cemeteries.Public | Public cemetery and crematorium facilities |
| 11 | Cemeteries.Other | Other cemetery and funeral service entities critical for public health and social continuity |
| 12 | Chemical.Production | Production of hazardous and high-consequence chemicals |
| 13 | Chemical.Storage | Chemical storage and distribution facilities |
| 14 | Chemical.Other | Other chemical sector entities critical for safety and security |
| 15 | Civil.ElectionMonitoring | Election observation and monitoring entities |
| 16 | Civil.HumanitarianAid | Humanitarian aid organizations |
| 17 | Civil.NGOs | Non-governmental organizations with critical functions |
| 18 | Civil.Other | Other civil society entities critical for democratic processes and social stability |
| 19 | Cultural.Archives | National and regional archives |
| 20 | Cultural.HistoricalSites | Historical monuments and archaeological sites |
| 21 | Cultural.Libraries | National and public libraries |
| 22 | Cultural.Museums | Museums and exhibition spaces |

| Rank | Keyword | Definition |
|---|---|---|
| 23 | Cultural.Other | Other cultural heritage entities critical for national identity and continuity |
| 24 | Defense.CommandControl | Military C4ISR systems (command, control, communications, computers, intelligence, surveillance, reconnaissance) |
| 25 | Defense.Cyber | Military cyber defense and security operations centers |
| 26 | Defense.IndustrialBase | Defense manufacturing, weapons systems, munitions production |
| 27 | Defense.Logistics | Military supply chains, fuel depots, ammunition storage |
| 28 | Defense.Installations | Military bases, headquarters, and operational facilities |
| 29 | Defense.Research | Defense laboratories and R&D facilities |
| 30 | Defense.Other | Other defense and military entities critical for national security |
| 31 | Digital.CloudServices | Cloud computing and critical digital services |
| 32 | Digital.DataCenters | Data center hosting and infrastructure services |
| 33 | Digital.DigitalProviders | Online marketplaces, search engines and social media platforms |
| 34 | Digital.DomainNameSystems | DNS service providers and TLD registries |
| 35 | Digital.ICTServiceManagement | ICT service management B2B services |
| 36 | Digital.SatelliteCommunications | Satellite communication networks and ground stations |
| 37 | Digital.TelecomNetworks | Fixed and mobile telecommunications networks |
| 38 | Digital.UnderseaCables | Submarine communications cable infrastructure |
| 39 | Digital.Other | Other digital and telecommunications entities critical for the continuity of digital services |
| 40 | Education.PrimarySecondary | Primary and secondary schools |
| 41 | Education.ResearchSchools | Research-focused educational institutions |
| 42 | Education.Universities | Universities and higher education institutions |
| 43 | Education.Other | Other educational entities critical for societal continuity |
| 44 | Emergency.CivilProtection | Civil protection and disaster management agencies |

| Rank | Keyword | Definition |
|---|---|---|
| 45 | Emergency.Medical | Ambulance services and emergency medical response |
| 46 | Emergency.FireAndRescue | Firefighting and rescue services |
| 47 | Emergency.Police | Law enforcement agencies and operations |
| 48 | Emergency.Other | Other emergency response entities critical for public safety |
| 49 | Energy.DistrictHeating | Operation of district heating networks |
| 50 | Energy.Electricity | Generation, transmission and distribution of electrical power |
| 51 | Energy.Gas | Transport, storage and distribution of natural gas |
| 52 | Energy.Hydrogen | Production, transport and storage of hydrogen |
| 53 | Energy.Nuclear | Nuclear power generation and fuel cycle facilities |
| 54 | Energy.Oil | Refining, transport and storage of petroleum products |
| 55 | Energy.Other | Other energy entities critical for the continuity of energy supply |
| 56 | Finance.MarketInfrastructures | Stock exchanges and clearing houses |
| 57 | Finance.Insurance | Systemically important insurance entities |
| 58 | Finance.PaymentSystems | Card schemes, transfers and instant payment systems |
| 59 | Finance.Other | Other financial entities critical for the stability of the financial system |
| 60 | Food.AgriculturalProduction | Strategic agricultural and livestock production |
| 61 | Food.FoodDistribution | Retail and logistics for food supply chains |
| 62 | Food.FoodProcessing | Industrial food processing and manufacturing |
| 63 | Food.FoodSafety | Food safety inspection and control authorities |
| 64 | Food.Other | Other food sector entities critical for the continuity of the food supply chain |
| 65 | Health.BloodAndTissue | Blood banks, tissue banks, and transplantation services |
| 66 | Health.HealthcareProviders | Hospitals and clinics providing essential care |
| 67 | Health.Laboratories | Medical analysis and diagnostic laboratories |
| 68 | Health.MedicalDevices | Manufacture and maintenance of critical medical devices |
| 69 | Health.PharmaceuticalSupplyChain | Manufacturing, wholesale and distribution of pharmaceuticals |
| 70 | Health.PublicHealth | Public health agencies and epidemiological surveillance |

| Rank | Keyword | Definition |
|---|---|---|
| 71 | Health.Other | Other health entities critical for the continuity of healthcare services |
| 72 | Logistics.FreightForwarding | Freight forwarding and cargo management |
| 73 | Logistics.LastMileDelivery | Last-mile delivery services |
| 74 | Logistics.ThirdPartyLogistics | Third-party logistics providers |
| 75 | Logistics.Warehousing | Strategic warehousing and storage facilities |
| 76 | Logistics.Other | Other logistics entities critical for supply chain continuity |
| 77 | Manufacturing.Aerospace | Aerospace and aviation manufacturing |
| 78 | Manufacturing.Batteries | Battery manufacturing and energy storage production |
| 79 | Manufacturing.Chemical | Chemical manufacturing and industrial processes |
| 80 | Manufacturing.Electronic | Manufacture of computer, electronic and optical products |
| 81 | Manufacturing.Defense | Defense manufacturing and strategic military supply chains |
| 82 | Manufacturing.MedicalDevices | Manufacture of medical devices and equipment |
| 83 | Manufacturing.MotorVehicles | Manufacture of motor vehicles and transport equipment |
| 84 | Manufacturing.Pharmaceutical | Pharmaceutical manufacturing and active pharmaceutical ingredients |
| 85 | Manufacturing.Semiconductors | Semiconductor fabrication and microelectronics manufacturing |
| 86 | Manufacturing.Strategic | Critical manufacturing such as metallurgy and components |
| 87 | Manufacturing.Other | Other manufacturing entities critical for industrial resilience and strategic supply chains |
| 88 | Media.Infrastructure | Transmitters, towers, and broadcast distribution infrastructure |
| 89 | Media.OnlineMedia | Digital news platforms and content providers |
| 90 | Media.Press | Newspapers, publishing houses, and press agencies |
| 91 | Media.Radio | Radio broadcasting networks and studios |
| 92 | Media.Television | Television broadcasting networks and studios |
| 93 | Media.Other | Other media entities critical for information integrity and public communication |
| 94 | Mining.CriticalRawMaterials | Extraction and processing of critical raw materials |
| 95 | Mining.EnergyMinerals | Coal, uranium, and other energy mineral mining |

| Rank | Keyword | Definition |
|---|---|---|
| 96 | Mining.Other | Other mining entities critical for resource security |
| 97 | Nuclear.FuelCycle | Nuclear fuel production, enrichment, reprocessing, and waste management |
| 98 | Nuclear.Medical | Nuclear medical facilities and radioisotope production |
| 99 | Nuclear.PowerGeneration | Civil nuclear power plants and associated facilities |
| 100 | Nuclear.Research | Nuclear research reactors and laboratories |
| 101 | Nuclear.Other | Other nuclear entities critical for safety and security |
| 102 | Postal.CourierServices | Postal and courier services |
| 103 | Postal.Other | Other postal and courier entities critical for the continuity of mail and parcel services |
| 104 | Religious.PilgrimageSites | Major pilgrimage destinations |
| 105 | Religious.PlacesOfWorship | Churches, mosques, synagogues, temples, and other religious buildings |
| 106 | Religious.Other | Other religious sites critical for community continuity |
| 107 | Research.BiologicalSafety | BSL-3 and BSL-4 high-containment laboratories |
| 108 | Research.ChemicalSafety | High-containment chemical research facilities |
| 109 | Research.Research | Key research laboratories with strategic importance |
| 110 | Research.Other | Other research entities critical for strategic research continuity and innovation |
| 111 | Space.GroundStations | Satellite ground control and telemetry stations |
| 112 | Space.LaunchFacilities | Space launch sites and associated infrastructure |
| 113 | Space.SpaceActivities | Space industry and satellite operations |
| 114 | Space.Other | Other space entities critical for the continuity of space-based services and infrastructure |
| 115 | Transport.Aviation | Airports, air traffic control and airline operations |
| 116 | Transport.Maritime | Ports, terminals and maritime traffic management |
| 117 | Transport.Pipeline | Oil, gas, and hydrogen pipeline infrastructure |
| 118 | Transport.PublicTransport | Urban and regional public transportation systems |
| 119 | Transport.Rail | Railway infrastructure and train operations |
| 120 | Transport.Road | Traffic management and strategic road logistics |

| Rank | Keyword | Definition |
|---|---|---|
| 121 | Transport.Other | Other transport entities critical for the continuity of passenger and freight mobility |
| 122 | Waste.Hazardous | Collection, treatment and disposal of hazardous waste |
| 123 | Waste.NonHazardous | Management of non-hazardous solid waste |
| 124 | Waste.Nuclear | Nuclear waste management and storage facilities |
| 125 | Waste.Recycling | Waste processing and recycling operations |
| 126 | Waste.Other | Other waste management entities critical for the continuity of waste services |
| 127 | Water.DamsAndReservoirs | Dam and reservoir infrastructure for water management |
| 128 | Water.DrinkingWater | Capture, treatment and distribution of potable water |
| 129 | Water.Irrigation | Large-scale agricultural irrigation systems |
| 130 | Water.Wastewater | Collection and treatment of sewage and wastewater |
| 131 | Water.Other | Other water entities critical for the continuity of water supply and sanitation |

Table 6: enum_alert_entitysector

## B.2. Alert Category

**uank;Keyword;Description**
0;Abuse.Coercion;"The
Ex:
1;Abuse.Extermism;"The
Ex:
2;Abuse.Grooming;"The
Ex:
3;Abuse.Harassment;"A
Ex:
4;Abuse.Trafficking;"The
Ex:
5;Abuse.Other;"Any
Ex:
6;Access.Authorized;"An
Ex:
7;Access.Backdoor;"The
Ex:
8;Access.Cloned;"The
Ex:
9;Access.Compromise;"An
Ex:
10;Access.Escalation;"An

**uank;Keyword;Description**
Ex:
11;Access.Forced;"An
Ex:
12;Access.Lost;"An
Ex:
13;Access.Tailgating;"A
Ex:
14;Access.Unauthorized;"Incident
Ex:
15;Access.Other;"Any
Ex:
16;Anomaly.Burst;"A
Ex:
17;Anomaly.Chaos;"A
Ex:
18;Anomaly.Context;"A
Ex:
19;Anomaly.Correlation;"An
Ex:
20;Anomaly.Deviation;"Any
Ex:
21;Anomaly.Drift;"A
Ex:
22;Anomaly.Oscillation;"A
Ex:
23;Anomaly.Relation;"A
Ex:
24;Anomaly.Sequence;"An
Ex:
25;Anomaly.Threshold;"Crossing
Ex:
26;Anomaly.Timing;"An
Ex:
27;Anomaly.Unknown;"An
Ex:
28;Anomaly.Other;"Any
Ex:
29;Availability.DDoS;"An
Ex:
30;Availability.DoS;"An
Ex:
31;Availability.Failure;"An
Ex:
32;Availability.Flapping;"A
Ex:
33;Availability.HeartBeat;"A
Ex:
34;Availability.Misconfiguration;"An
Ex:
35;Availability.Outage;"An
Ex:
36;Availability.Overload;"An
Ex:

**uank;Keyword;Description**
37;Availability.Recovery;"The
Ex:
38;Availability.Restart;"The
Ex:
39;Availability.Start;"The
Ex:
40;Availability.Status;"A
Ex:
41;Availability.Stop;"The
Ex:
42;Availability.Other;"Any
Ex:
43;Biological.Animal;"An
Ex:
44;Biological.Epidemic;"The
Ex:
45;Biological.Insect;"An
Ex:
46;Biological.Zombies;"A
Ex:
47;Biological.Other;"Any
Ex:
48;Climat.Drought;"A
Ex:
49;Climat.LakeOutburst;"An
Ex:
50;Climat.Wildfire;"An
Ex:
51;Climat.Other;"Any
Ex:
52;Extraterrestrial.Aliens;"A
Ex:
53;Extraterrestrial.Impact;"An
Ex:
54;Extraterrestrial.SpaceWeather;"An
Ex:
55;Extraterrestrial.Other;"Any
Ex:
56;Fraud.Copyright;"The
Ex:
57;Fraud.Corruption;"A
Ex:
58;Fraud.Espionage;"The
Ex:
59;Fraud.Masquerade;"A
Ex:
61;Fraud.Usage;"The
Ex:
62;Fraud.Other;"Any
Ex:
63;Geophysical.Earthquake;"An
Ex:
64;Geophysical.MassMovement;"An

**uank;Keyword;Description**
Ex:
66;Geophysical.Volcanic;"An
Ex:
65;Geophysical.Other;"Any
Ex:
67;Hydro.Flood;"An
Ex:
68;Hydro.Landslide;"An
Ex:
69;Hydro.Wave;"An
Ex:
70;Hydro.Other;"Any
Ex:
71;Insider.Malicious;"A
Ex:
72;Insider.Negligent;"A
Ex:
73;Insider.Other;"Any
Ex:
74;Malware.Adware;"A
Ex:
75;Malware.Backdoor;"A
Ex:
76;Malware.Cryptominer;"A
Ex:
77;Malware.Downloader;"A
Ex:
78;Malware.Ransomware;"A
Ex:
79;Malware.Rootkit;"A
Ex:
80;Malware.Spyware;"A
Ex:
81;Malware.Trojan;"A
Ex:
82;Malware.Virus;"A
Ex:
83;Malware.Worm;"A
Ex:
84;Malware.Other;"Any
Ex:
85;Meteo.Cold;"A
Ex:
86;Meteo.Fog;"A
Ex:
87;Meteo.Heat;"A
Ex:
88;Meteo.Rain;"A
Ex:
89;Meteo.Snow;"A
Ex:
90;Meteo.Wind;"A
Ex:

**uank;Keyword;Description**
91;Meteo.Other;"Any
Ex:
92;National.Conflict;"A
Ex:
93;National.Crime;"Large-scale
Ex:
94;National.Cyber;"Large-scale
Ex:
95;National.Economical;"An
Ex:
96;National.Environmental;"Incidents
Ex:
97;National.Societal;"An
Ex:
98;National.Terrorism;"An
Ex:
99;National.Other;"Any
Ex:
100;Operational.Configuration;"A
Ex:
101;Operational.Misuse;"The
Ex:
102;Operational.PolicyViolation;"An
Ex:
103;Operational.ProcessFailure;"An
Ex:
104;Operational.Update;"The
Ex:
105;Operational.Other;"Any
Ex:
106;Recon.Aerial;"The
Ex:
107;Recon.Landscape;"The
Ex:
108;Recon.Network;"The
Ex:
109;Recon.OSINT;"The
Ex:
110;Recon.Other;"Any
Ex:
111;Sabotage.Data;"The
Ex:
112;Sabotage.Destruction;"The
Ex:
113;Sabotage.Disruption;"The
Ex:
114;Sabotage.Equipment;"The
Ex:
115;Sabotage.Graffiti;"The
Ex:
116;Sabotage.Tampering;"The
Ex:
117;Sabotage.Vandalism;"The

**uank;Keyword;Description**
Ex:
118;Sabotage.Other;"Any
Ex:
119;Safety.Accident;"An
Ex:
120;Safety.Aggression;"An
Ex:
121;Safety.Explosion;"A
Ex:
122;Safety.Fire;"An
Ex:
123;Safety.Hostage;"An
Ex:
124;Safety.Sexual;"An
Ex:
125;Safety.Other;"Any
Ex:
126;SocialEng.Baiting;"A
Ex:
127;SocialEng.Phishing;""""A
128;SocialEng.Pretexting;"A
Ex:
129;SocialEng.QuidProQuo;"A
Ex:
133;SocialEng.Other;"Any
Ex:
134;SupplyChain.Compromise;"A
Ex:
135;SupplyChain.Disruption;"An
Ex:
136;SupplyChain.Other;"Any
Ex:
137;Theft.Breaches;"A
Ex:
138;Theft.Data;"The
Ex:
139;Theft.Equipment;"The
Ex:
140;Theft.FinInfo;"The
Ex:
141;Theft.IP;"The
Ex:
142;Theft.Machinery;"The
Ex:
143;Theft.PII;"The
Ex:
144;Theft.Other;"Any
Ex:
145;Other.Uncategorised;"Any
Ex:
146;Other.Undetermined;"An
Ex:
147;Other.Test;"An

**uank;Keyword;Description**
Ex:
148;Other.ext-value;"A
Ex:

Table 7: enum_alert_category

## B.3. Analyzer Category

| uank | Keyword | Description |
|---|---|---|
| 0 | Undetermined | The analyzer category has not been determined, or cannot be established from available information. Ex: an analyzer whose type is unknown at ingestion time. |
| 1 | APP.BAST | Bastion Host — A hardened, intermediary server that provides secure remote access to a protected network, acting as a single controlled entry point. Ex: an SSH jump host, a RDP gateway. |
| 2 | APP.DAST | Dynamic Application Security Testing — Testing a running application from the outside to find security weaknesses in its runtime behaviour, without access to source code. Ex: a scanner probing a live web app for injection flaws. |
| 3 | APP.IAST | Interactive Application Security Testing — Hybrid testing that instruments a running application from within (agent-based) while it is exercised, combining runtime and code-level visibility. Ex: an agent inside a Java app reporting tainted data flows. |
| 4 | APP.RASP | Runtime Application Self-Protection — Security embedded inside a running application that detects and blocks attacks in real time from within the runtime, independently of the surrounding infrastructure. Ex: an in-app module blocking a SQL injection attempt. |
| 5 | APP.SANDBOX | Sandbox / File Detonation — Isolated execution environment that analyses the behaviour of suspicious files or URLs |

| uank | Keyword | Description |
|---|---|---|
| | | and decides whether to raise an alert. Ex: a suspicious attachment detonated in a VM to observe its behaviour. |
| 6 | APP.SAST | Static Application Security Testing — Analysis of application source code or binaries without executing them, to find vulnerabilities early in development. Ex: a linter flagging unsanitised input in source. |
| 7 | APP.SCA | Software Composition Analysis — Analysis of third-party dependencies and libraries for known vulnerabilities, licence issues, and supply-chain risk. Ex: detecting a vulnerable open-source component in a build. |
| 8 | CLD.CASB | Cloud Access Security Broker — A control point between users and cloud services that enforces security policy, visibility, and compliance for cloud usage. Ex: blocking unsanctioned SaaS uploads. |
| 9 | CLD.CIEM | Cloud Infrastructure Entitlement Management — Management and least-privilege enforcement of identities, roles, and permissions across cloud environments. Ex: detecting over-permissive IAM roles. |
| 10 | CLD.CSPM | Cloud Security Posture Management — Continuous monitoring of cloud configurations for misconfigurations, drift, and compliance violations. Ex: flagging a publicly exposed storage bucket. |
| 11 | CLD.CWPP | Cloud Workload Protection Platform — Security for cloud workloads (VMs, containers, serverless) covering runtime, vulnerability, and threat detection. Ex: detecting a cryptominer in a container. |
| 12 | DDoS.ANTI-DDOS | Distributed Denial of Service Protection — Systems and techniques that detect and mitigate volumetric or |

| uank | Keyword | Description |
|---|---|---|
| | | application-layer DDoS attacks. Ex: rate-limiting and scrubbing malicious floods. |
| 13 | DDoS.SCRUB | Scrubber / Scrubbing Center — A dedicated traffic-cleaning facility that diverts and filters attack traffic before forwarding clean traffic to the target. Ex: a carrier scrubbing centre absorbing a 1 Tbps flood. |
| 14 | DDoS.WAF-DDOS | Web Application Firewall with DDoS Protection — A WAF that also provides DDoS mitigation for HTTP/HTTPS traffic. Ex: blocking a Layer 7 HTTP flood against a web service. |
| 15 | EMAIL.ANTI-PHISH | Anti-Phishing — Detection and blocking of phishing attempts in email, including malicious links, spoofed senders, and deceptive content. Ex: quarantining a credential-harvesting email. |
| 16 | EMAIL.DMARC | Domain-based Message Authentication, Reporting and Conformance — Email authentication (SPF/DKIM/ DMARC) monitoring and enforcement to prevent sender spoofing. Ex: rejecting mail failing DMARC alignment. |
| 17 | EMAIL.SEG | Secure Email Gateway — A comprehensive email security appliance/service filtering inbound and outbound mail for spam, malware, phishing, and policy violations. Ex: a gateway scanning attachments before delivery. |
| 18 | EMAIL.SPAM-FILTER | Spam Filter — Detection and filtering of unsolicited bulk email based on content, reputation, and heuristics. Ex: moving a bulk advertisement to the junk folder. |
| 19 | END.AM | Application Allowlisting — Execution control that permits only explicitly approved applications to run, blocking all others by default. Ex: a workstation that only runs signed corporate apps. |

| uank | Keyword | Description |
|---|---|---|
| 20 | END.AV | Antivirus — Signature- and heuristic-based detection and removal of known malware on endpoints. Ex: quarantining a file matching a known virus signature. |
| 21 | END.DLP-EP | Endpoint Data Loss Prevention — Prevention of sensitive data leakage from endpoint devices via channels such as USB, clipboard, print, or email. Ex: blocking a copy of customer data to a USB stick. |
| 22 | END.EDR | Endpoint Detection and Response — Continuous endpoint monitoring, behavioural detection, and response capabilities for advanced threats. Ex: isolating a host showing lateral-movement behaviour. |
| 23 | END.EPP | Endpoint Protection Platform — A consolidated endpoint security suite combining antivirus, firewall, EDR, and other protections in a single agent. Ex: a single agent providing AV, host firewall, and threat hunting. |
| 24 | END.HIDS | Host Intrusion Detection System — Detection of malicious or anomalous activity on a host by monitoring system calls, files, and logs. Ex: alerting on a suspicious process spawning from a web server. |
| 25 | END.HIPS | Host Intrusion Prevention System — Active prevention of host-level intrusions by blocking malicious system activity in real time. Ex: blocking an unauthorised process from modifying system files. |
| 26 | END.HPT | Honeypot — A decoy system or service designed to attract and detect attackers, providing early warning and intelligence. Ex: a fake SSH server logging brute-force attempts. |
| 27 | HUM.REPORT | Human Report — A human observer who analyses a situation and decides to raise |

| uank | Keyword | Description |
|---|---|---|
| | | an alert, acting as the analyzer in the IDMEFv2 sense. Ex: a guard reporting a suspicious crowd approaching a building, an operator qualifying an incident in a control room, a patrol officer signalling an anomaly. |
| 28 | ID.DCAP | Data-Centric Audit and Protection — Monitoring and auditing of data access and usage, focused on the data itself rather than the infrastructure. Ex: alerting on unusual access to a sensitive database. |
| 29 | ID.DLP | Data Loss Prevention — Detection and prevention of sensitive data exfiltration across all channels (network, endpoint, cloud). Ex: blocking an outbound email containing PII. |
| 30 | ID.IAM | Identity and Access Management — Governance of identities and enforcement of access control across systems and applications. Ex: provisioning, authenticating, and authorising user accounts. |
| 31 | ID.IRM | Identity Risk Management — Assessment and scoring of identity-related risk, such as dormant accounts, excessive privileges, or compromised credentials. Ex: flagging a service account with no owner. |
| 32 | ID.MFA | Multi-Factor Authentication Analyzer — Analysis of multi-factor authentication events to detect bypass attempts, repeated failures, or anomalous challenges. Ex: alerting on repeated MFA push notifications ignored by a user. |
| 33 | ID.PAM | Privileged Access Management — Control, vaulting, and monitoring of privileged accounts and sessions. Ex: a vault checking out an admin password for a session. |
| 34 | ID.PIM | Privileged Identity Management — Just-in-time and lifecycle management |

| uank | Keyword | Description |
|---|---|---|
| | | of privileged identities and roles. Ex: granting temporary admin rights for a maintenance window. |
| 35 | ID.SECRETS | Secrets Detection — Analysis of code repositories, configurations, and logs to detect exposed secrets such as API keys, tokens, or certificates. Ex: detecting an API key committed to a public repository. |
| 36 | ID.UEBA | User and Entity Behaviour Analytics — Behavioural analytics over users and entities to detect anomalies and insider threats. Ex: alerting on a user downloading an unusual volume of files. |
| 37 | NAT.AIR | Air Quality Sensor — Analysis of airborne particles and gases to detect pollution, smoke, or hazardous releases. Ex: detecting a pollution peak or a smoke plume from a fire. |
| 38 | NAT.FIRE | Wildfire Detection — Detection of uncontrolled fires in natural or rural areas, typically via thermal or infrared sensing. Ex: a thermal sensor detecting a wildfire starting in a forest. |
| 39 | NAT.HYDRO | Hydrological Sensor — Analysis of water levels, flow, and quality to detect floods, droughts, or contamination. Ex: a river gauge alerting on a rising water level. |
| 40 | NAT.METEO | Meteorological Sensor — Analysis of atmospheric conditions (temperature, wind, precipitation, pressure) to detect weather events. Ex: a weather station alerting on a heatwave or storm. |
| 41 | NAT.RADIO | Radiation Sensor — Analysis of ionizing radiation levels to detect nuclear incidents or radioactive releases. Ex: a radiation monitor alerting on abnormal levels near a facility. |
| 42 | NAT.SAT | Earth Observation Satellite — Analysis of satellite imagery to detect large-scale natural |

| uank | Keyword | Description |
|---|---|---|
| | | events, environmental changes, or disasters. Ex: satellite imagery detecting a cyclone, a wildfire, or a flood. |
| 43 | NAT.SEISMO | Seismic Sensor — Analysis of ground motion to detect earthquakes, volcanic activity, or subsurface events. Ex: a seismograph alerting on a magnitude 6.0 earthquake. |
| 44 | NET.DNS-FW | DNS Firewall — Filtering of DNS queries to block resolution of malicious or unwanted domains. Ex: blocking resolution of a known C2 domain. |
| 45 | NET.DPI | Deep Packet Inspection — Inspection of packet payloads beyond headers to identify protocols, applications, or malicious content. Ex: detecting malware signatures inside HTTPS traffic. |
| 46 | NET.FW | Firewall — Filtering and policy enforcement of network traffic based on addresses, ports, and protocols. Ex: blocking inbound traffic to a database port from the internet. |
| 47 | NET.NAC | Network Access Control — Enforcement of endpoint compliance and identity before granting network access. Ex: quarantining a non-patched laptop until it is remediated. |
| 48 | NET.NBAD | Network Behaviour Anomaly Detection — Detection of deviations from normal network behaviour patterns, typically via baselining and statistics. Ex: alerting on an unusual spike in DNS queries from one host. |
| 49 | NET.NDR | Network Detection and Response — Continuous network monitoring, behavioural detection, and response for advanced threats. Ex: correlating east-west traffic to identify lateral movement. |
| 50 | NET.NGFW | Next-Generation Firewall — A firewall with application awareness, user identity, and integrated threat prevention |

| uank | Keyword | Description |
|---|---|---|
| | | (IPS, AV, URL filtering). Ex: allowing only sanctioned apps regardless of port. |
| 51 | NET.NIDS | Network Intrusion Detection System — Passive monitoring of network traffic to detect known attack patterns and anomalies. Ex: alerting on a signature matching an exploit attempt. |
| 52 | NET.NIPS | Network Intrusion Prevention System — Inline inspection of network traffic that can actively block detected attacks. Ex: dropping packets matching an exploit signature. |
| 53 | NET.PROXY | Proxy Server — An intermediary for client requests that enforces ACLs and can inspect TLS sessions. Ex: a forward proxy logging and filtering outbound web traffic. |
| 54 | NET.WAF | Web Application Firewall — Filtering of HTTP/ HTTPS traffic to protect web applications from common attacks (OWASP Top 10, etc.). Ex: blocking a SQL injection attempt against a web app. |
| 55 | NET.WIDS | Wireless Intrusion Detection System — Passive monitoring of wireless networks to detect rogue APs, evil twins, and wireless attacks. Ex: alerting on an unauthorised AP in the building. |
| 56 | NET.WIPS | Wireless Intrusion Prevention System — Active prevention of wireless threats, including containment of rogue devices. Ex: deauthenticating clients from a rogue AP. |
| 57 | OT.IOT-IDS | IoT Intrusion Detection System — Detection of threats targeting IoT devices, whose constraints (power, compute, protocols) differ from general IT. Ex: alerting on anomalous MQTT traffic from a sensor. |
| 58 | OT.OT-IDS | Operational Technology IDS — Detection of threats in industrial control systems (SCADA, DCS), where |

| uank | Keyword | Description |
|---|---|---|
| | | availability and safety dominate. Ex: alerting on unauthorised Modbus writes to a PLC. |
| 59 | OT.PLC-SCAN | PLC Scanner — Discovery and vulnerability assessment of PLC/SCADA devices in operational environments. Ex: enumerating PLCs on a plant network and checking firmware versions. |
| 60 | OT.SCADA | SCADA Monitor — Analysis of supervisory control and data acquisition systems to detect abnormal process states or operational anomalies. Ex: alerting on an abnormal pressure reading in an industrial process. |
| 61 | OT.SENSOR | Industrial Sensor — Analysis of industrial measurements (pressure, temperature, flow) to detect abnormal process conditions. Ex: a flow sensor alerting on a sudden drop in a pipeline. |
| 62 | PHY.1D-LAS | Laser Sensor 1D — Measures distance or presence along a single axis using a laser beam. Ex: a laser rangefinder, a beam-break tripwire. |
| 63 | PHY.2D-LAS | Laser Sensor 2D — Scans a plane with a laser to detect presence, contours, or movement in two dimensions. Ex: a planar laser scanner guarding an entrance. |
| 64 | PHY.3D-LAS | Laser Sensor 3D — Scans a volume with a laser to produce a three-dimensional representation of a scene. Ex: a 3D scanner producing a point cloud of a room. |
| 65 | PHY.ACCESS-CTRL | Access Control System — Physical entry/exit control using credentials (badge, PIN, biometric) and monitoring of door state. Ex: a badge reader logging who enters a server room. |
| 66 | PHY.ADS | Anti-Drone System — Detection (and sometimes neutralisation) of unauthorised |

| uank | Keyword | Description |
|---|---|---|
| | | drones via RF, radar, or acoustic sensing. Ex: detecting a drone approaching a restricted site. |
| 67 | PHY.CBRN | CBRN Detector — Detection of chemical, biological, radiological, or nuclear agents in the environment. Ex: a detector alerting on a chemical agent release. |
| 68 | PHY.FIRE | Fire Detector — Detection of heat, flame, or combustion products indicating a fire. Ex: a flame detector triggering an alarm in a warehouse. |
| 69 | PHY.FR-CAM | Face Recognition Camera — A camera coupled with biometric software that identifies or verifies individuals from facial features. Ex: matching a person against an authorised list at a gate. |
| 70 | PHY.GAS | Gas Detector — Detection of hazardous or combustible gases in the environment. Ex: a methane detector alerting in a confined space. |
| 71 | PHY.GLASS-BRK | Glass Break Detector — An acoustic sensor detecting the specific sound signature of breaking glass. Ex: triggering an alarm when a window is smashed. |
| 72 | PHY.HAR | Human Activity Recognition — AI-based analysis of video or sensor data to recognise human activities and behaviours. Ex: detecting a person falling, running, or loitering. |
| 73 | PHY.LWIR | Long-Wave Infrared — Thermal imaging in the long-wave IR band (roughly 8–14 µm), suited to ambient-temperature detection. Ex: a thermal camera detecting body heat at night. |
| 74 | PHY.MOT-SEN | Motion Sensor — A sensor (PIR, microwave, ultrasonic) detecting movement in its field of view. Ex: a PIR sensor turning on a light and logging presence. |

| uank | Keyword | Description |
|---|---|---|
| 75 | PHY.MWIR | Mid-Wave Infrared — Thermal imaging in the mid-wave IR band (roughly 3–5 µm), used for higher-temperature targets and some gas detection. Ex: a MWIR camera monitoring an industrial furnace. |
| 76 | PHY.OBJ-DET | Object Detection Camera — A camera with AI models that detect and classify objects in the scene. Ex: detecting a left bag or a vehicle in a restricted zone. |
| 77 | PHY.PERIMETER | Perimeter Detection — Detection of intrusion across a perimeter using barriers, fibre optics, or microwave sensing. Ex: a fence-mounted fibre sensor alerting on an attempted climb. |
| 78 | PHY.RADAR | Radar Sensor — Detection of movement, presence, or objects using radio waves. Ex: a ground radar detecting an intrusion across a perimeter. |
| 79 | PHY.SMOKE | Smoke Detector — Detection of smoke particles indicating a fire. Ex: a smoke detector triggering an alarm in a server room. |
| 80 | PHY.SWIR | Short-Wave Infrared — Imaging in the short-wave IR band (roughly 0.9–1.7 µm), useful for seeing through haze and some materials. Ex: a SWIR camera inspecting contents through a container. |
| 81 | PHY.VAD | Voice Activity Detection — Audio analysis that detects the presence of human speech in a signal. Ex: triggering recording only when speech is present. |
| 82 | PHY.VNIR | Visible and Near-Infrared — Multi-spectral imaging covering the visible and near-IR bands. Ex: a VNIR camera used in low-light surveillance. |
| 83 | PHY.WATER | Water Leak Detector — Detection of water presence or leakage in a monitored area. Ex: a leak sensor alerting in a data centre. |

| uank | Keyword | Description |
|---|---|---|
| 84 | SIEM.APM | Application Performance Monitoring — Analysis of application performance and availability to detect degradations, errors, or outages. Ex: alerting on a spike in response time for a critical service. |
| 85 | SIEM.ETL | Extract, Transform, Load — Data pipeline tooling that collects, normalises, and routes data to downstream stores. Ex: Logstash, Fluentd, Vector. |
| 86 | SIEM.LOG | Log Analyzer — Aggregation, indexing, and search of log data for troubleshooting and security analysis. Ex: an ELK stack or Splunk index. |
| 87 | SIEM.NMS | Network Management System — Monitoring and management of network devices, links, and performance. Ex: a NMS alerting on a down interface. |
| 88 | SIEM.SIEM | Security Information and Event Management — Centralised collection, correlation, and alerting across security-relevant data sources. Ex: correlating firewall and endpoint events into one alert. |
| 89 | SIEM.SOAR | Security Orchestration, Automation and Response — Automation of security workflows and incident response playbooks. Ex: auto-enriching an alert and opening a ticket. |
| 90 | SIEM.UPTIME | Uptime Monitoring — Analysis of service and infrastructure availability to detect outages and recoveries. Ex: alerting when a critical service becomes unreachable. |
| 91 | TI.CTI | Cyber Threat Intelligence — Strategic and operational intelligence about threats, actors, and campaigns, produced for decision-making. Ex: a report on an APT targeting your sector. |
| 92 | TI.TI-FEED | Threat Intelligence Feed — A stream of structured threat data (IOCs, TTPs) consumed for |

| uank | Keyword | Description |
|---|---|---|
| | | detection and enrichment. Ex: a STIX/TAXII feed of malicious IPs. |
| 93 | TI.TIP | Threat Intelligence Platform — Aggregation, correlation, and management of threat intelligence from multiple sources. Ex: a platform correlating IOCs from several feeds. |
| 94 | VM.ASM | Attack Surface Management — Discovery and continuous monitoring of an organisation's externally exposed assets. Ex: finding an unknown public subdomain. |
| 95 | VM.PENTEST | Penetration Testing Tools — Tools supporting manual or automated offensive testing of systems and applications. Ex: exploitation frameworks and scanners used by testers. |
| 96 | VM.VULN-SCANNER | Vulnerability Scanner — Automated assessment of systems and applications for known vulnerabilities. Ex: a scanner reporting missing patches and CVEs. |
| 97 | ext-value | A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1) Ex: a vendor-specific analyzer type declared through the extension mechanism. |

Table 8: enum_analyzer_category

## B.4. Analyzer Data

| Rank | Item | Description |
|---|---|---|
| 0 | Undetermined | The analyzed data type is unknown or not yet classified. Used as a placeholder when no other value applies. Example: a generic analyzer whose input could not be identified. |
| 1 | Acoustics | Sound pressure or specific frequency measurements captured by a microphone or acoustic sensor. Example: gunshot detection, glass break |

| Rank | Item | Description |
|---|---|---|
| | | detection, machine noise spectrum. |
| 2 | Alert | IDMEFv2 or equivalent alert produced by an upstream analyzer and consumed as input by a correlator, aggregator or hypervisor. Example: IDMEFv2 JSON alert from a local SOC processed by a central SOC correlator. |
| 3 | Contact | Physical interaction or binary state change detected by a contact sensor. Example: door contact switch, cabinet open detection, tamper switch. |
| 4 | Content | Payload or body of a communication inspected at the application layer. Example: HTTP body, email message, SMTP DATA section, file transfer payload, DPI-matched bytes. |
| 5 | Data | Generic or opaque data stream whose structure is not further specified. Example: raw sensor telemetry, vendor-specific binary stream, serial output. |
| 6 | Datagram | Raw packet-level data captured at the network layer. Example: Ethernet frame, IPv4/IPv6 packet, TCP segment, UDP datagram, ICMP message. |
| 7 | External | Data originating from a third party such as a threat intelligence feed or an external CERT. Example: STIX/TAXII feed, IOC list, external CERT notification, vendor advisory. |
| 8 | File | Content or metadata of a file read by the analyzer for inspection, integrity or change detection. Example: file hash, PE header, document content, file permissions, modification timestamp. |
| 9 | Flow | Network flow records aggregating packets by 5-tuple or similar criteria. Example: NetFlow, IPFIX, sFlow records exported by routers or probes. |
| 10 | Fog | Visibility reduction or optical density measured by a fog sensor. Example: fog detection |

| Rank | Item | Description |
|---|---|---|
| | | on a perimeter road, visibility monitoring for cameras. |
| 11 | Humidity | Relative humidity or moisture content in the air measured by a hygrometer. Example: datacenter humidity level, warehouse moisture monitoring. |
| 12 | Images | Visible-spectrum image or video frames captured by a camera. Example: CCTV frame, IP camera stream, facial recognition input. |
| 13 | Lidar | Point cloud or distance measurements produced by a laser-based sensor. Example: 3D point cloud from a rotating lidar, perimeter scanning lidar. |
| 14 | Light | Ambient light level or flicker detected by a photodetector. Example: lux measurement, room lighting state, strobe or flicker detection. |
| 15 | Log | Structured or unstructured event records produced by systems, applications or security devices. Example: syslog messages, Windows EventLog entries, journald records, firewall logs, application traces. |
| 16 | Magnetic | Magnetic field strength or anomaly detected by a magnetometer. Example: door open/close magnetic contact, vehicle detection, magnetic anomaly near equipment. |
| 17 | Particles | Concentration of dust, smoke or airborne contaminants measured by a particle sensor. Example: smoke detector, PM2.5 sensor, dust monitoring in a facility. |
| 18 | PCAP | Packet capture files stored on disk and analyzed offline. Example: .pcap and .pcapng files produced by tcpdump, dumpcap or Wireshark. |
| 19 | Rain | Precipitation detected or measured by a rain sensor. Example: rainfall detection on an outdoor site, rain gauge. |
| 20 | Seismic | Ground motion or earthquake-related signals captured by |

| Rank | Item | Description |
|---|---|---|
| | | a seismograph. Example: earthquake detection, ground vibration near a datacenter. |
| 21 | SNMP | Data retrieved via the Simple Network Management Protocol from managed devices. Example: SNMP GET/WALK results, MIB object values, traps and informs. |
| 22 | Temperature | Ambient, device or surface temperature measured by a thermometer or thermal probe. Example: server room temperature, device CPU temperature, cold-chain monitoring. |
| 23 | Thermal | Infrared image or heat signature data captured by a thermal sensor. Example: thermal camera frame, heat map of a room or equipment. |
| 24 | Vibration | Mechanical oscillation or structural movement measured by an accelerometer or vibration sensor. Example: seismic-grade vibration of a server rack, structural health monitoring of a bridge. |
| 25 | Water | Presence, flow or level of water detected by a sensor. Example: leak detection under a raised floor, flood sensor, water flow meter. |
| 26 | ext-value | A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1) |

Table 9: enum_analyzer_data

## B.5. Analyzer Method

| Rank | Keyword | Description |
|---|---|---|
| 0 | Undetermined | Analyser method is undetermed |
| 1 | AI | An analyzer that uses machine learning, deep learning, or other artificial intelligence techniques to learn normal behavior and detect sophisticated or novel threats. |
| 2 | Anomaly | An analyzer that identifies deviations from established |

| Rank | Keyword | Description |
|---|---|---|
| | | norms or baselines without relying on predefined signatures, often flagging unusual patterns in traffic, behavior, or system activity. |
| 3 | Behavioral | An analyzer that monitors and analyzes the actions of users, entities, or processes over time to detect malicious or suspicious activities that deviate from expected behavior patterns. |
| 4 | Biometric | An analyzer that uses electronic devices to capture and measure unique physical or behavioral characteristics (e.g., fingerprint, iris, voice) for identification or authentication purposes. |
| 5 | Blackhole | A method that analyses traffic destined for a non-existent or sinkhole route to identify malicious activity, such as connections to known command-and-control servers or scanning from infected hosts. |
| 6 | Contextual | An analyzer that enriches raw events with additional context (e.g., asset value, user role, time of day, business criticality) to prioritize alerts and assess true impact. |
| 7 | Correlation | An analyzer that aggregates and examines multiple, disparate data streams or events to identify complex relationships, patterns, sequences, or dependencies that indicate a security incident. |
| 8 | Ensemble | An analyzer that combines multiple detection methods (e.g., signature, anomaly, behavioral) to improve accuracy, reduce false positives, and detect threats that single methods might miss. |
| 9 | Fingerprinting | An analyzer that creates unique identifiers or "fingerprints" for devices, applications, or network stacks to detect spoofing, unauthorized devices, or configuration changes. |
| 10 | Frequency | An analyzer that detects incidents based on the rate |

| Rank | Keyword | Description |
|---|---|---|
| | | or regularity of events (e.g., repeated failed logins, rapid-fire requests) exceeding or falling below expected frequencies. |
| 11 | Fusion | An analyzer that combines data from multiple heterogeneous sensors and sources to create a comprehensive, high-confidence view of an incident, reducing ambiguity and false alerts. |
| 12 | Geolocation | An analyzer that determines the physical location of an asset or event (e.g., login attempt, IP address, device) and flags activities occurring from unexpected or high-risk locations. |
| 13 | Graph-based | An analyzer that models relationships between entities (users, devices, processes) as graphs and detects anomalies or attack paths by analyzing connections, dependencies, and traversals. |
| 14 | Heat | An analyzer (sensor or device) that detects, measures, and monitors thermal energy (infrared radiation) to identify anomalies like fires, overheating equipment, or human presence. |
| 15 | Heuristic | An analyzer that detects potentially unknown threats by using algorithmic logic, rules of thumb, or suspicious characteristics rather than relying on specific signature matches. |
| 16 | Honeypot | A decoy system or resource designed to lure, detect, and analyze malicious activity by mimicking a legitimate target, diverting attackers away from real assets. |
| 17 | Hygiene | An analyzer that continuously checks systems and configurations against security best practices, compliance standards, or hardening guidelines to identify weaknesses or drift. |

| Rank | Keyword | Description |
|---|---|---|
| 18 | Integrity | An analyzer that monitors critical system components (files, configurations, registry keys) for unauthorized changes, verifying their integrity against a known good baseline. |
| 19 | Metadata | An analyzer that examines the data about data (e.g., file creation timestamps, email headers, connection logs) to uncover hidden relationships or suspicious attributes. |
| 20 | Monitor | An analyzer that continuously observes a system, network, or environment to track its state, health, or activity, often providing real-time alerts on specific conditions. |
| 21 | Movement | An analyzer (sensor or system) that detects, tracks, and quantifies physical motion using technologies like radar, lidar, or video analytics. |
| 22 | Orchestration | An analyzer that coordinates and triggers automated response actions based on detected incidents, often integrated with SOAR (Security Orchestration, Automation, and Response) platforms. |
| 23 | Pattern | An analyzer that identifies specific sequences, combinations, or recurring arrangements of events or data that indicate malicious activity, even if individual elements appear benign. |
| 24 | Policy | An analyzer that evaluates events, configurations, or behaviors against a set of predefined rules, configurations, or compliance requirements to detect violations or misconfigurations. |
| 25 | Predictive | An analyzer that uses historical data and modeling to forecast potential future incidents, vulnerabilities, or attack vectors before they occur. |
| 26 | Protocol | An analyzer that validates network traffic or communications against expected protocol |

| Rank | Keyword | Description |
|---|---|---|
| | | specifications, RFC compliance, or standard behavior to detect anomalies or malicious variations. |
| 27 | Recon | An analyzer that actively or passively probes or monitors an environment to discover assets, services, or vulnerabilities, often as part of a defensive assessment or adversary simulation. |
| 28 | Reputation | An analyzer that evaluates the trustworthiness of an entity (e.g., IP address, domain, file hash) by checking it against known threat intelligence lists, blocklists, or reputation scores. |
| 29 | Rule-based | An analyzer that applies conditional logic (if-then-else statements) defined by experts to correlate events and generate alerts based on specific combinations of conditions. |
| 30 | Sequence | An analyzer that detects threats by examining the order and timing of events, identifying attack chains or kill chain progressions (e.g., scan # exploit # installation # C2). |
| 31 | Signature | An analyzer that detects known threats by matching events or patterns against a database of specific signatures, hashes, or Indicators of Compromise (IoCs). |
| 32 | Statistical | An analyzer that detects anomalies by establishing a baseline of normal behavior and identifying events that deviate significantly from expected statistical parameters. |
| 33 | Tarpit | A mechanism that intentionally slows down or delays suspicious connections (e.g., network connections or login attempts) to hinder automated attacks and scanning. |
| 34 | Threat Intelligence | An analyzer that ingests and matches internal events against external threat feeds, IoC lists, and adversary TTPs (Tactics, |

| Rank | Keyword | Description |
|---|---|---|
| | | Techniques, and Procedures) to identify known threats. |
| 35 | Threshold | An analyzer that detects incidents by comparing a metric or count (e.g., number of failed logins, traffic volume) against a predefined limit or threshold. |
| 36 | Trend | An analyzer that monitors data over extended periods to identify gradual changes, emerging patterns, or long-term shifts that may indicate evolving threats or security degradation. |
| 37 | ext-value | A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1) |

Table 10: Analyzer Methods

## B.6. Source Category

| Rank | Keyword | Description |
|---|---|---|
| 0 | Undetermined | Source Category is undetermined. |
| 1 | Acoustic.Sound.GlassBreak | Specific frequencies (glass breaking, gunshots) |
| 2 | Acoustic.Sound.Infrasound | Infrasound (nuclear explosions, volcanic activity) |
| 3 | Acoustic.Sound.PressurizedLeak | Leaks in pressurized pipes (high-frequency hiss) |
| 4 | Acoustic.Sound.SonarEcho | Sonar echoes (submarines, fish, seafloor mapping) |
| 5 | Acoustic.Sound.UltrasonicNoise | Ultrasonic noise (for pest detection or leak detection) |
| 6 | Acoustic.Vibration.Footstep | Footsteps (geophones or seismic sensors) |
| 7 | Acoustic.Vibration.Seismic | Seismic activity (earthquakes, tremors) |
| 8 | Acoustic.Vibration.StructuralCrack | Structural integrity (cracks via acoustic emission) |
| 9 | Acoustic.Vibration.VehicleVibe | Vehicle movement (magnetic or seismic) |
| 10 | Acoustic.Vibration.VibAnalysis | Machinery imbalance (vibration analysis) |
| 11 | ChemBio.Gases.CO | Carbon Monoxide (CO) |
| 12 | ChemBio.Gases.CWA | CWA (Chemical Warfare Agents - Nerve gas, Blister agents) |
| 13 | ChemBio.Gases.Freon | Refrigerant leaks (Freon) |
| 14 | ChemBio.Gases.Hydrogen | Hydrogen |

| Rank | Keyword | Description |
|---|---|---|
| 15 | ChemBio.Gases.Methane | Natural Gas / Methane |
| 16 | ChemBio.Gases.O2Level | Oxygen levels (deficiency or enrichment) |
| 17 | ChemBio.Gases.Propane | Propane / LPG |
| 18 | ChemBio.Gases.Radon | Radon |
| 19 | ChemBio.Gases.VOC | Volatile Organic Compounds (VOCs) - paint fumes, off-gassing |
| 20 | ChemBio.Properties.ORP | Oxidation-Reduction Potential (ORP) |
| 21 | ChemBio.Properties.pHLevel | pH levels (acidity/alkalinity) |
| 22 | ChemBio.Properties.Salinity | Salinity |
| 23 | ChemBio.Threats.Anthrax | Anthrax spores |
| 24 | ChemBio.Threats.Botulinum | Botulinum toxin |
| 25 | ChemBio.Threats.Pathogen | Pathogens in water (E. coli, Cholera) |
| 26 | ChemBio.Threats.Ricin | Ricin |
| 27 | Cyber.Application.SQLQuery | Malicious code injected into a database query (SQL Injection). |
| 28 | Cyber.Application.XSSPayload | The script injected into a webpage to steal data or redirect users. |
| 29 | Cyber.Cloud.APIToken | A compromised key used to access cloud resources (AWS, Azure) or SaaS platforms. |
| 30 | Cyber.Cloud.BucketName | The specific cloud storage location where data was stored or exfiltrated from. |
| 31 | Cyber.Cloud.CloudResourceID | The unique ID of the compromised virtual machine or database. |
| 32 | Cyber.Cloud.OAuthApp | A malicious third-party app granted permissions to a user's data. |
| 33 | Cyber.Email.EmailAuthResult | The authentication result showing if the email passed security checks. |
| 34 | Cyber.Email.MessageID | A unique identifier for the email used to track it across mail servers. |
| 35 | Cyber.Email.RecipientEmail | The target email address of the attack. |
| 36 | Cyber.Email.ReplyToEmail | The address where replies go; often different from the sender for tracking. |
| 37 | Cyber.Email.ReturnPath | The address where non-delivery receipts go; used in spoofing analysis. |
| 38 | Cyber.Email.SenderEmail | The "From" address used in a phishing campaign. |
| 39 | Cyber.Email.XOrigIP | The original IP address of the sender (if recorded by the mail server). |

| Rank | Keyword | Description |
|---|---|---|
| 40 | Cyber.Endpoint.CmdLine | The exact command line used to execute the malware; often reveals attacker intent. |
| 41 | Cyber.Endpoint.DLL | The name of a malicious library loaded into a legitimate process (DLL Sideloading). |
| 42 | Cyber.Endpoint.FileHash | The cryptographic fingerprint of the malicious file (SHA256, MD5). |
| 43 | Cyber.Endpoint.FilePath | The location on the file system where the malware resides. |
| 44 | Cyber.Endpoint.MemoryArtifact | Code or processes that exist only in RAM (volatile memory) and not on disk. |
| 45 | Cyber.Endpoint.PID | The unique identifier of the malicious process at the time of analysis. |
| 46 | Cyber.Endpoint.ProcessName | The name of the malicious executable file running on the endpoint. |
| 47 | Cyber.Endpoint.RegistryKey | The Windows Registry path used for persistence or configuration changes. |
| 48 | Cyber.Endpoint.ScheduledTask | The name of a task created by the attacker for persistence or execution. |
| 49 | Cyber.Endpoint.ServiceName | The name of a malicious service installed to run the malware. |
| 50 | Cyber.Forensics.NetFlow | Metadata about the connection (who talked to whom, when, and data volume). |
| 51 | Cyber.Forensics.PacketPayload | The raw data being transmitted containing passwords or exfiltrated data. |
| 52 | Cyber.Hardware.DeviceSerial | The unique identifier of a specific compromised laptop or mobile device. |
| 53 | Cyber.Hardware.USBDeviceID | The specific make/model/ serial of a USB drive used to introduce malware. |
| 54 | Cyber.Identity.LoginTimestamp | The specific time of an unusual login (e.g., 3:00 AM from a foreign location). |
| 55 | Cyber.Identity.Username | The specific user account that was compromised or used for lateral movement. |
| 56 | Cyber.Network.IPv4 | The source IP of an attacker, a Command & Control (C2) server, or a malicious scan. |
| 57 | Cyber.Network.IPv6 | The source IP in IPv6 format; increasingly used to bypass IPv4-based allowlists. |

| Rank | Keyword | Description |
|---|---|---|
| 58 | Cyber.Network.MAC | The physical device identifier on a local network; useful if the attacker is on-site. |
| 59 | Cyber.Network.Port | The specific source or destination port used (e.g., 445 for ransomware spreading). |
| 60 | Cyber.Network.Protocol | The network protocol used; anomalous use may indicate tunneling (e.g., DNS tunneling). |
| 61 | Cyber.Web.Domain | The domain hosting malware or Command & Control infrastructure (e.g., evil.com). |
| 62 | Cyber.Web.HTTPMethod | Unusual HTTP methods like PUT or DELETE used to manipulate data. |
| 63 | Cyber.Web.ReferrerHeader | The page the user was on before clicking the malicious link. |
| 64 | Cyber.Web.SessionID | A hijacked session token used to impersonate a user without a password. |
| 65 | Cyber.Web.URIPath | The specific directory or file requested on the web server. |
| 66 | Cyber.Web.URL | The full malicious link pointing to a payload or phishing page. |
| 67 | Cyber.Web.UserAgent | The browser or application signature used by the attacker when connecting. |
| 68 | ElectroMag.Infrared.GasLeakIR | Gas leaks (specific gases absorb specific IR wavelengths) |
| 69 | ElectroMag.Infrared.HeatTemp | Heat (temperature of objects) |
| 70 | ElectroMag.Infrared.MoistureContent | Moisture content (in materials) |
| 71 | ElectroMag.Infrared.ThermalGradient | Thermal gradients (heat loss in buildings) |
| 72 | ElectroMag.Infrared.WarmBody | Warm bodies (humans, animals) via Passive Infrared (PIR) |
| 73 | ElectroMag.Ultraviolet.BioContamUV | Biological contamination |
| 74 | ElectroMag.Ultraviolet.CoronaArc | Corona discharge (electrical arcing) |
| 75 | ElectroMag.Ultraviolet.FlameUV | Flames (fire detection) |
| 76 | ElectroMag.Ultraviolet.ForgeryUV | Forgery (detecting UV security features on documents) |
| 77 | ElectroMag.Light.BiometricEye | Biological signatures (retinas, irises) - Failed recognition indicates intrusion attempt |
| 78 | ElectroMag.Light.EdgesShapes | Edges and shapes (for machine vision) |
| 79 | ElectroMag.Light.Flicker | Flicker (from faulty lighting or machinery) |
| 80 | ElectroMag.Light.LightColor | Color and wavelength (spectral content) |

| Rank | Keyword | Description |
|---|---|---|
| 81 | ElectroMag.Light.LightPresence | Presence of light (ambient brightness) |
| 82 | ElectroMag.Light.MotionOptical | Motion (optical flow, changes in pixel arrays) |
| 83 | ElectroMag.Light.ObjectsGeneric | Specific objects (via computer vision: faces, vehicles, text, barcodes) |
| 84 | Force.Atmospheric.Altitude | Altitude (in aircraft) |
| 85 | Force.Atmospheric.BaroPress | Barometric pressure (weather changes) |
| 86 | Force.Atmospheric.VacuumLevel | Vacuum levels (in manufacturing) |
| 87 | Force.Fluid.Airflow | Airflow (differential pressure) |
| 88 | Force.Fluid.WaterPressure | Water pressure (for leak detection via pressure drop) |
| 89 | Force.Touch.GripStrength | Grip strength |
| 90 | Force.Touch.PhysicalContact | Physical contact (bumpers on robots, touch screens) |
| 91 | Force.Touch.TirePressure | Tire pressure |
| 92 | Force.Touch.Weight | Weight (scales, load cells) |
| 93 | Radiation.General.Alpha | Alpha particles |
| 94 | Radiation.General.Beta | Beta particles |
| 95 | Radiation.General.Cosmic | Cosmic rays |
| 96 | Radiation.General.Gamma | Gamma rays |
| 97 | Radiation.General.Neutron | Neutron radiation |
| 98 | Radiation.General.RadonDecay | Radon gas (decay products) |
| 99 | Radiation.General.Xray | X-rays |
| 100 | Magnetic.General.CurrentFlow | Current flow (Hall effect sensors, clamp meters) |
| 101 | Magnetic.General.ESD | Electrostatic discharge |
| 102 | Magnetic.General.Ferrous | Ferrous metals (via magnetic disturbance) |
| 103 | Magnetic.General.MagneticAnomaly | Magnetic anomalies (submarines, buried structures) |
| 104 | Magnetic.General.Orientation | Orientation (Compass / Magnetometer - detecting Earth's magnetic field) |
| 105 | Magnetic.General.Powerline | Power lines (AC magnetic fields) |
| 106 | Magnetic.General.Voltage | Voltage (non-contact voltage testers) |
| 107 | Object.Indoor.Container | Bottles / Cups / Cutlery |
| 108 | Object.Indoor.Document | Books / Papers / Documents |
| 109 | Object.Indoor.Electronics | Electronics (laptops / phones / TVs) |
| 110 | Object.Indoor.Furniture | Furniture (chairs / tables / sofas) |
| 111 | Object.Indoor.Weapon | Weapons (guns / knives) |
| 112 | Object.LivingBeings.Animal | Animals (general wildlife/ livestock/pets) |
| 113 | Object.LivingBeings.BodyPose | Pose (skeleton keypoints / body position) |
| 114 | Object.LivingBeings.EyeGaze | Eyes (gaze tracking / eye blink) |

| Rank | Keyword | Description |
|---|---|---|
| 115 | Object.LivingBeings.Face | Faces (facial recognition) |
| 116 | Object.LivingBeings.HandGesture | Hands (hand gestures / raising hand) |
| 117 | Object.LivingBeings.Human | Humans (person/people detection) |
| 118 | Object.Outdoor.Building | Buildings / Houses |
| 119 | Object.Outdoor.Pothole | Potholes / Road cracks |
| 120 | Object.Outdoor.RoadSign | Road signs / Traffic lights |
| 121 | Object.Outdoor.Vegetation | Trees / Plants / Vegetation |
| 122 | Object.SpecificScenes.FireVisual | Fire / Flames (visual spectrum) |
| 123 | Object.SpecificScenes.Flood | Flooding / Water accumulation |
| 124 | Object.SpecificScenes.SmokeVisual | Smoke plume (visual spectrum) |
| 125 | Object.SpecificScenes.Snow | Snow coverage |
| 126 | Object.Vehicles.Aircraft | Aircraft (drones / planes / helicopters) |
| 127 | Object.Vehicles.Boat | Boats / Ships |
| 128 | Object.Vehicles.Bus | Buses |
| 129 | Object.Vehicles.Car | Cars / Automobiles |
| 130 | Object.Vehicles.LicensePlate | License plates (Automatic Number Plate Recognition - ANPR) |
| 131 | Object.Vehicles.Moto | Motorcycles / Bicycles |
| 132 | Object.Vehicles.Truck | Trucks |
| 133 | Particulate.Aerosols.Bioaerosol | Viral or bacterial particles (bio-aerosol detectors) |
| 134 | Particulate.Aerosols.Dust | Airborne dust (particulate counters) |
| 135 | Particulate.Aerosols.Pollen | Pollen and allergens |
| 136 | Particulate.Aerosols.SmokeIonization | Combustion particles (Ionization smoke detectors) |
| 137 | Particulate.Aerosols.SmokePhotoelectric | Visible smoke obscuration (Photoelectric smoke detectors) |
| 138 | Particulate.Liquids.ChemicalSpill | Chemical spills (specific conductivity or pH changes) |
| 139 | Particulate.Liquids.FuelSheen | Fuel/Oil sheen on water |
| 140 | Particulate.Liquids.LiquidLevel | Level of liquid in a tank (ultrasonic, capacitive, float) |
| 141 | Particulate.Liquids.WaterLeak | Water leakage (conductivity or float switches) |
| 142 | Particulate.Solids.DensityChange | Density changes (in manufacturing) |
| 143 | Particulate.Solids.ForeignObject | Foreign objects in food (X-ray machines) |
| 144 | Particulate.Solids.GrainFlow | Grain flow (in agriculture) |
| 145 | Particulate.Solids.Landmine | Landmines/UXO (Ground Penetrating Radar) |
| 146 | Particulate.Solids.Metal | Metal (Metal detectors via eddy currents) |
| 147 | Software.Detected.Crowd | Crowd formation |
| 148 | Software.Detected.CyberAttack | Cyber-attack (Anomalous network traffic patterns) |
| 149 | Software.Detected.Drowsiness | Drowsiness (Eye closure detection via camera) |

| Rank | Keyword | Description |
|---|---|---|
| 150 | Software.Detected.Explosion | Explosion (Sudden light flash + pressure wave + sound) |
| 151 | Software.Detected.FailedBiometric | Failed retina/iris scan (multiple failures may indicate intrusion) |
| 152 | Software.Detected.FireFusion | Fire (Combination of smoke, heat, and flicker) |
| 153 | Software.Detected.Intrusion | Intrusion (Crossing a virtual line) |
| 154 | Software.Detected.Loitering | Loitering (Person staying too long) |
| 155 | Software.Detected.Tailgating | Tailgating (Following someone through a door) |
| 156 | ext-value | A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1) |

Table 11: Source Categories

## B.7. Target Category

| Rank | Keyword | Description |
|---|---|---|
| 0 | Undetermined | Target Category is undetermined |
| 1 | Commercial.ClientBase | Client portfolio, active contracts |
| 2 | Commercial.Partnerships | Strategic alliances, partner networks |
| 3 | Commercial.Reputation | Brand image, online reputation, goodwill |
| 4 | Financial.Cash | Liquidities, bank accounts |
| 5 | Financial.Investments | Shares, bonds, holdings in subsidiaries |
| 6 | Financial.Receivables | Customer receivables, promissory notes |
| 7 | Furniture.Fittings | Lighting, partitions, flooring |
| 8 | Furniture.HVAC | Air conditioning, heating systems (non-industrial) |
| 9 | Furniture.OfficeFurniture | Desks, chairs, filing cabinets |
| 10 | Human.EmployerBrand | Employment contracts, recruitment appeal |
| 11 | Human.Organization | Work processes, internal know-how |
| 12 | Human.Skills | Employee expertise, team competencies |
| 13 | Infrastructure.CivilWorks | Bridges, loading docks, quays |
| 14 | Infrastructure.Foundations | Special foundations, internal roads, pavements |
| 15 | Infrastructure.Networks | Buried water, gas, electricity, fiber networks |
| 16 | IP.Copyrights | Software code, manuals, artistic works |

| Rank | Keyword | Description |
|---|---|---|
| 17 | IP.Patents | Registered patents, inventions |
| 18 | IP.Trademarks | Brand names, logos, registered designs |
| 19 | IT.Endpoints | Computers, printers, scanners |
| 20 | IT.Hardware | Servers, storage arrays, cabling |
| 21 | IT.Telecom | WiFi access points, phones, smartphones |
| 22 | Logistics.Containers | Containers, trailers, wagons |
| 23 | Logistics.SpecialVehicles | Mobile cranes, aerial work platforms |
| 24 | Logistics.Vehicles | Cars, trucks, forklifts |
| 25 | Marketing.Content | Market studies, ad campaigns, social media |
| 26 | Marketing.Domains | Website URLs, domain names |
| 27 | Marketing.Identity | Visual identity, logo, brand guidelines |
| 28 | Production.HeavyEquipment | Presses, furnaces, compressors, generators |
| 29 | Production.Machinery | Assembly lines, industrial robots, conveyors |
| 30 | Production.MachineTools | Lathes, milling machines, industrial 3D printers |
| 31 | RealEstate.Buildings | Factories, warehouses, offices, commercial premises |
| 32 | RealEstate.Land | Unbuilt land, forests, or agricultural grounds |
| 33 | RealEstate.SpecialStructures | Hangars, silos, parking lots |
| 34 | Software.Applications | ERP, CRM, SaaS platforms |
| 35 | Software.Data | Customer databases, supplier lists, prospects |
| 36 | Software.Licenses | Software usage rights, subscriptions |
| 37 | ext-value | A value used to indicate that this attribute is extended and the actual value is provided using the corresponding ext-* attribute. (see Section 4.1.1) |

Table 12: Target Categories

# Appendix C. Examples

This section contains several examples of events/incidents which may be described using the IDMEF Data Model defined in.

For each example, the serialization method listed in Section 5 was used on the original IDMEF message to produce a JSON representation.

## C.1. Physical intrusion

Listing 1 describes an incident where an unidentified man was detected on company premises near the building where server room A is located.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b1",
  "Description": "Potential intruder detected",
  "Type": "Physical",
  "Priority": "Low",
  "Status": "Incident",
  "Cause": "Malicious",
  "CreateTime": "2021-05-10T16:52:13.075994+00:00",
  "StartTime": "2021-05-10T16:52:13+00:00",
  "Category": [
    "Access.Forced"
  ],
  "Analyzer": {
    "Name": "BigBrother",
    "Hostname": "bb.acme.com",
    "Model": "Big Brother v42",
    "Category": [
      "PHY.HAR",
      "FRC.FR-CAM"
    ],
    "Data": [
      "Images"
    ],
    "Method": [
      "Movement",
      "Biometric",
      "AI"
    ],
    "IP": "192.0.2.1"
  },
  "Sensor": [
    {
      "IP": "192.0.2.2",
      "Name": "Camera #23",
      "Model": "SuperDuper Camera v1",
      "Location": "Hallway to server room A1"
    }
  ],
  "Source": [
    {
      "Note": "Black Organization, aka. APT 4869"
    }
  ],
  "Attachment": [
    {
      "Name": "wanted",
      "FileName": "fbi-wanted-poster.jpg",
      "Size": 1234567,
      "Ref": ["https://www.fbi.gov/wanted/topten"],
      "ContentType": "image/jpg",
      "ContentEncoding": "base64",
      "Content": "..."
    },
    {
      "Name": "pic01",
      "Note": "Hi-res picture showing John Doe near server room A1",
      "ExternalURI": ["ftps://192.0.2.1/cam23/20210510165211.jpg"],
      "ContentType": "image/jpg"
    }
```

## C.2. Cyberattack

Listing 2 describes an incident related to a potential bruteforce attack against the "root" user account of the server at 192.0.2.2 and 2001:db8::/32.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b2",
  "Description": "Potential bruteforce attack on root user account",
  "Type": "Cyber",
  "Priority": "Medium",
  "CreateTime": "2021-05-10T16:55:29.196408+00:00",
  "StartTime": "2021-05-10T16:55:29+00:00",
  "Category": [
    "Access.Forced"
  ],
  "Analyzer": {
    "Name": "SIEM",
    "Hostname": "siem.acme.com",
    "Model": "Concerto SIEM 5.2",
    "Category": [
      "SIEM.SIEM",
      "SIEM.LOG"
    ],
    "Data": [
      "Log"
    ],
    "Method": [
      "Monitor",
      "Signature"
    ],
    "IP": "192.0.2.1"
  },
  "Sensor": [
    {
      "IP": "192.0.2.5",
      "Name": "syslog",
      "Hostname": "www.acme.com",
      "Model": "rsyslog 8.2110",
      "Location": "Server room A1, rack 10"
    }
  ],
  "Target": [
    {
      "IP": "192.0.2.2",
      "Hostname": "www.acme.com",
      "Location": "Server room A1, rack 10",
      "User": "root"
    },
    {
      "IP": "2001:db8::/32",
      "Hostname": "www.acme.com",
      "Location": "Server room A1, rack 10",
      "User": "root"
    }
  ]
}
```

## C.3. Server outage

Listing 3 describes an incident where the webserver at "www.example.com" encountered some kind of failure condition resulting in an outage.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b3",
  "Description": "A server did not reply to an ICMP ping request",
  "Type": "Availability",
  "Priority": "Medium",
  "Status": "Incident",
  "Cause": "Unknown",
  "CreateTime": "2021-05-10T16:59:11.875209+00:00",
  "StartTime": "2021-05-10T16:59:11.875209+00:00",
  "Category": [
    "Availability.Outage"
  ],
  "Analyzer": {
    "Name": "NMS",
    "Hostname": "nms.example.com",
    "Model": "Concerto NMS 5.2",
    "Category": [
      "SIEM.NMS"
    ],
    "Data": [
      "Network"
    ],
    "Method": [
      "Monitor"
    ],
    "IP": "192.0.2.1"
  },
  "Target": [
    {
      "IP": "192.168.1.2",
      "Hostname": "www.acme.com",
      "Service": "website",
      "Location": "Server room A1, rack 10"
    }
  ]
}
```

## C.4. Combined incident

Listing 4 describes a combined incident resulting from the correlation of the previous physical, cyber and availability incidents.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b4",
  "Description": "Intrusion and Sabotage detected",
  "Type": "Combined",
  "Priority": "High",
  "Status": "Incident",
  "Cause": "Malicious",
  "CreateTime": "2021-05-10T16:59:15.075994+00:00",
  "StartTime": "2021-05-10T16:52:11+00:00",
  "Category": [
    "Access.Unauthorized",
    "Sabotage.Data",
  ,
  "CorrelID": [
    "819df7bc-35ef-40d8-bbee-1901117370b1",
    "819df7bc-35ef-40d8-bbee-1901117370b2",
    "819df7bc-35ef-40d8-bbee-1901117370b3"
  ],
  "Analyzer": {
    "Name": "Correlator",
    "Hostname": "correlator.acme.com",
    "Model": "Concerto Hybrid Correlator v5.2",
    "Category": [
    ],
    "Data": [
      "Alert"
    ],
    "Method": [
      "Correlation"
    ],
    "IP": "192.0.2.1"
  },
  "Source": [
    {
      "Note": "Black Organization, aka. APT 4869"
    }
  ],
  "Target": [
    {
      "Location": "Server room A1"
    },
    {
      "IP": "192.0.2.2",
      "Hostname": "www.acme.com",
      "User": "root"
    },
    {
      "IP": "192.0.2.2",
      "Hostname": "www.acme.com",
      "Service": "website"
    }
  ]
}
```

## C.5. Hazard incident

Listing 5 describes a heavy snow storm announced in 48h on Paris and Versailles.

```
{
  "Version": "2.D.V07",
  "ID": "819df7bc-35ef-40d8-bbee-1901117370b1",
  "Description": "Snow storm forecast",
  "Priority": "Low",
  "Status": "Incident",
  "Cause": "Hazard",
  "Confidence": 0.8,
  "CreateTime": "2021-05-10T16:52:13.075994+00:00",
  "StartTime": "2021-05-12T10:00:00+00:00",
  "EndTime": "2021-05-12T10:00:00+00:00",
  "Type": ["Physical"],
  "Category": [
    "Meteo.Snow",
    "Meteo.Wind",
    "Meteo.Cold"
  ],
  "Analyzer": {
    "Name": "Weather Monitor",
    "Hostname": "weather.acme.com",
    "IP": "192.0.2.1"
  },
  "Source": [
    {
      "Note": "Heavy snow storm coming from North"
    }
  ],
  "Target": [
    {
      "GeoLocation": "48.8584,2.2945",
      "UnLocation": "FR PAR",
      "Location": "Acme Paris Site"
    },
    {
      "GeoLocation": "48.8019,2.1301",
      "UnLocation": "FR VER",
      "Location": "Acme Versailles Site"
    }
  ]
}
```

# Appendix D. JSON Validation Schema (Non-normative)

Listing 5 contains a JSON Schema that can be used to validate incoming IDMEF messages prior to processing. Please note that extraneous linebreaks have been included due to formatting constraints.

```
{
    "description": "JSON schema for the Intrusion Detection Message
 Exchange Format (IDMEF) version 2 (revision 2.D.V08)",
    "properties": {
        "Version": {
            "description": "The version of the IDMEF format in use by this
 alert. During the drafts tuning period the version is equal to the draft
 version. Therefore it is \"2.D.V0X\" for Draft V0X.",
            "enum": [
                "2.D.V08"
            ]
        },
        "ID": {
            "description": "Unique identifier for the alert.",
            "$ref": "#/definitions/uuidType"
        },
        "OrganisationName": {
            "description": "Corporate/Main Office Organisation Name Useful
 if alerts are sent to a multi-organisation central incident detection
 manager. Example: ACME Corporation",
            "type": "string"
        },
        "OrganisationId": {
            "description": "Corporate/Main Office Organisation ID. Where
 possible official organisation ID manage by national authority. Useful
 if alerts are sent to a multi-organisation central incident detection
 manager. This ID has to be chosen depending on the overall detection
 perimeter and the nature of the monitored organisation (Private/Public,
 Commercial, International, etc.) Examples: OrganisationId in France could
 be SIREN, in England could be CR, Germany could be Handelsregisternummer,
 Spain could be CIF, Italia could be Partita IVA, USA could be EIN, etc.
 Commercial OrganisationId in Europe could be V.A.T ID",
            "type": "string"
        },
        "EntityName": {
            "description": "Entity Name, monitored by the organisation,
 where the incident occurred. Could be a town, region or country name
 or an internal name. Could also be the name of a client for a MSSP
 centralizing it's client incidents in a single system. Do not repeat
 the organisation name in the EntityName Example: - ACME HeadQuaters
 is located in Paris France and has a local office in India - If the
 incident occurred in the local office: \"OrganisationName\": \"ACME\" ,
 \"EntityName\": \"India\" - If the incident occurred in the headquaters:
 \"OrganisationName\": \"ACME\", \"EntityName\": \"Headquaters\" (or
 \"Paris\")",
            "type": "string"
        },
        "EntityId": {
            "description": "Entity ID, monitored by the organisation,
 where the incident occurred. Useful if organisation and entity are not
 directly linked, like a client and a MSSP.",
            "type": "string"
        },
        "EntitySector": {
            "description": "The economic sector(s) and sub-sector(s)
 in which the entity operates. Values follow the dot notation
 sector.subsector based on the critical infrastructure taxonomy defined
 in the NIS2 Directive and CER (Critical Entities Resilience) Directive.
 This attribute enables sector-based correlation, regulatory compliance
 reporting, and risk context for incident detection.",
```

## Author's Address

**Gilles Lehmann**
Telecom SudParis
FR
EMail: gilles.lehmann@telecom-sudparis.eu